Theoretical MHD Modelling of the Polomac Concept

Ideal MHD and Equilibrium Derivation and a Framework for Numerical Implementation

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Abbreviations

CFL Courant-Friedrichs-Lewy

FDTD Finite difference time domain

HEI Hot-electron interchange

ITER International Thermonuclear Experimental Reactor

LDX Levitating Dipole Experiment

MHD Magnetohydrodynamic

PDE Partial differential equation

RT-1 Ring-trap 1

1. Introduction

1.1. Fusion as an Energy Source and the Confinement Challenge

As global energy demand continues to rise, fusion remains an attractive candidate for carbon-free baseload power. In magnetic confinement fusion, the challenge is no longer merely reaching fusion conditions, but sustaining stable, reactor-relevant plasma performance. The dominant approach relies on toroidal magnetic confinement in tokamaks and stellarators. For both geometries, great engineering challenges on the way to net energy gain and commercial power plants remain.

1.2. Poloidal Confinement and the Polomac Concept

Poloidal magnetic confinement has been explored in the past, most notably the by the Levitated Dipole Experiment (LDX) and also by the Japanese Ring-Trap 1 (RT-1) experiment. Due to promising initial results, the concept is now being revisited by the OpenStar project and the Polomac, proposed by Deutelio, which represents a new approach in this class. Its key engineering innovation is the use of magnetic tunnels, regions where external coils deflect outboard field lines aside, to support, feed, and cool the central dipole coil without direct plasma contact. As of this writing, the concept is at the design stage. The small Deutelio prototype is described in Elio et al. 2024, but no experimental plasma data have been published.

1.3. Scope and Structure of this Work

This work develops a theoretical MHD framework for the Polomac from first principles. Chapter 2 derives the ideal MHD equations, chapter 3 specializes them to the Polomac geometry, chapter 4 formulates the static equilibrium via a reduced Grad-Shafranov equation, chapter 5 evaluates the relevant dimensionless parameters, chapter 6 compares the Polomac to the tokamak and stellarator, and chapter 7 establishes a framework for a linearized FDTD solver. Chapter 8 concludes the thesis by summarizing the results, discussing future work, and acknowledging those who contributed to this project.
The thesis and accompanying documentation are publicly available at `https://github.com/nrupf/polomac-mhd`.

2. Single Fluid Ideal MHD Equations

This theory follows mostly Freidberg's book *Ideal MHD* (Freidberg 2014), Chen's book *Introduction to Plasma Physics and Controlled Fusion* (Chen 2016) and EPFL's *Introduction to Plasma Phyiscs* course (Fasoli et al. 2018).

2.1. Maxwell's Equations

The microscopic Maxwell equations are

$$\text{Gauss's law:} \quad \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}\,, \tag{2.1}$$

$$\text{Gauss's law for magnetism:} \quad \nabla \cdot \vec{B} = 0\,, \tag{2.2}$$

$$\text{Faraday's law of induction:} \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}\,, \tag{2.3}$$

$$\text{Ampère–Maxwell law:} \quad \nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}\,. \tag{2.4}$$

2.2. The Boltzmann Equation

The Boltzmann equation is

$$\frac{\partial f_s}{\partial t} + \vec{v}\nabla_{\vec{r}} f_s + \frac{q_s}{m_s}(\vec{E}^{\mathrm{lr}} + \vec{v} \times \vec{B}^{\mathrm{lr}})\nabla_{\vec{v}} f_s - \left(\frac{\partial f}{\partial t}\right)_{\mathrm{coll}} = 0\,.$$

Here the short range interactions are accounted for in the collisional term, and the long range (lr) interactions from collective effects are accounted for in the electric and Lorentz term. The critical screening distance hereby is the Debye length λ_D of electrons (Chen 2016)

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{e^2 n_e}}\,. \tag{2.5}$$

Acknowledging the above, we from now on omit the superscript *lr* and write simply $\vec{E}$ and $\vec{B}$.

2.2.1. Neglection of Gravity

The gravitational force on a single ion is approximately $F_G = mg \lesssim 1 \times 10^{-26}\,\mathrm{kg} \cdot 9.81\,\mathrm{ms}^{-2} \sim 1 \times 10^{-25}\,\mathrm{N}$. From Poisson's equation ($\nabla^2 \phi = \frac{\rho}{\epsilon_0}$) and a length scale

estimate for long range forces ($\nabla \sim \frac{1}{\lambda_D}$), we can estimate the long-range (beyond λ_D) Coulomb force to be $F_C = eE = e\nabla\phi \sim e\lambda_D \frac{\rho}{\epsilon_0} = \frac{e^2 n_e \lambda_D}{\epsilon_0} \sim 3 \times 10^{-27}\,\mathrm{m^2N} n_e \lambda_D$.
Even a very weak estimate ($\lambda_D \sim 10^{-6}$, $n_e \sim 10^{14}$) , compared to typical plasma values (see Blandford and Thorne 2013), this shows that gravity can be neglected.

2.3. Species Continuity Equation and Momentum Equation

2.3.1. Species Variables

- Number density: $n_s(\vec{r}, t) = \int d\vec{v}\, f_s(\vec{r}, \vec{v}, t)$

In general, we note the per particle average in velocity space of a function $g(\vec{v})$ at a given $(\vec{r}, t)$ as

$$\langle g(\vec{r}, t)\rangle_{\vec{v}} = \frac{1}{n_s(\vec{r}, t)} \int d\vec{v}\, g(\vec{v}) f_s(\vec{r}, \vec{v}, t)\,.$$

Thus we have

- Average fluid velocity $\vec{u}_s(\vec{r}, t)$:

$$g(\vec{v}) = \vec{v} \implies \vec{u}_s(\vec{r}, t) = \frac{1}{n_s(\vec{r}, t)} \int d\vec{v}\, \vec{v} f_s(\vec{r}, \vec{v}, t)\,.$$

- Average kinetic energy density $w_s(\vec{r}, t)$:

$$g(\vec{v}) = \frac{1}{2} m_s \vec{v}^2 \implies w_s(\vec{r}, t) = \frac{1}{n_s(\vec{r}, t)} \int d\vec{v}\, \frac{1}{2} m_s \vec{v}^2 f_s(\vec{r}, \vec{v}, t)\,.$$

Furthermore, we define the pressure tensor $\overleftrightarrow{P}$ as (note that we do not normalize over number density)

$$\overleftrightarrow{P}_s := m_s \int d\vec{v}\, (\vec{v} - \vec{u}_s) \otimes (\vec{v} - \vec{u}_s) f_s(\vec{r}, \vec{v}, t)\,.$$

We then evaluate moments of the Boltzmann equation

$$\int d\vec{v}\, g_s(\vec{v})\, [\text{Boltzmann}] = 0\,.$$

2.3.2. Species Continuity Equation

In order to obtain the species continuity equation, we choose $g_s(\vec{v}) = 1$ and obtain

$$\int d\vec{v} \left[\underbrace{\frac{\partial f_s}{\partial t}}_{(1)} + \underbrace{\vec{v} \cdot \frac{\partial f_s}{\partial \vec{r}}}_{(2)} + \underbrace{\frac{q_s}{m_s}(\vec{E} + \vec{v} \times \vec{B}) \cdot \frac{\partial f_s}{\partial \vec{v}}}_{(3)} - \underbrace{\left(\frac{\partial f}{\partial t}\right)_{coll}}_{(4)} \right] = 0\,,$$

where

$$(1): \quad \int d\vec{v}\,\frac{\partial f_s}{\partial t} = \frac{\partial}{\partial t}\int f_s = \frac{\partial}{\partial t} n_s\,,$$

$$(2): \quad \int d\vec{v}\,\vec{v}\cdot\frac{\partial f_s}{\partial \vec{r}} = \frac{\partial}{\partial \vec{r}}\cdot\int d\vec{v}\,\vec{v} f_s = \frac{\partial}{\partial \vec{r}}\cdot(n_s\vec{u}_s) \qquad \vec{v}\text{ indep. of }\vec{r}\,,$$

$$\begin{aligned}(3): \quad \int d\vec{v}\,\frac{q_s}{m_s}(\vec{E}+\vec{v}\times\vec{B})\cdot\frac{\partial f_s}{\partial \vec{v}} &= \frac{q_s}{m_s}\int\frac{\partial}{\partial \vec{v}}\cdot(\vec{E}+\vec{v}\times\vec{B})f_s \\ &= \frac{q_s}{m_s}\int_{S_{\vec{v}\to\infty}}(\vec{E}+\vec{v}\times\vec{B})\,f_s\,d\vec{S}_{\vec{v}} \\ &= 0 \quad \text{assuming } f_s \overset{v\to\infty}{\to} 0\,,\end{aligned}$$

$$(4): \quad \int d\vec{v}\left(\frac{\partial f}{\partial t}\right)_{coll} = 0 \quad \text{because we assume collisions don't create or destroy particles.}$$

We thus arrive at the species fluid continuity equation

$$\frac{\partial}{\partial t}n_s + \frac{\partial}{\partial \vec{r}}\cdot(n_s\vec{u}_s) = 0\,.$$

2.3.3. Species Momentum Equation

For the species momentum equation we choose $g_s(\vec{v}) = m_s\vec{v}$. A detailed derivation is attached in section A.1, the result we obtain is

$$m_s n_s\frac{d\vec{u}_s}{dt} = q_s n_s\left(\vec{E}+\vec{u}_s\times\vec{B}\right) - \frac{\partial}{\partial \vec{r}}\cdot\overleftrightarrow{P}_s + \vec{R}_s\,, \tag{2.6}$$

where $\frac{d}{dt}$ is the convective derivative, $\frac{d}{dt} = \frac{\partial}{\partial t} + \vec{u}_s\cdot\frac{\partial}{\partial \vec{r}}$.
$\vec{R}_s$ is a viscous force due to collisions, $\vec{R} = \int d\vec{v}\,m_s(\vec{v}-\vec{u}_s)\left(\frac{\partial f}{\partial t}\right)_{coll}$.
In the relaxation time approximation we have $\vec{R}_e \approx -m_e n_e \nu_{ei}(\vec{u}_e-\vec{u}_i)$, where ν_{ei} is the electron-ion collision frequency.

2.3.4. Species Energy Equation

For the energy equation we choose $g_s(\vec{v}) = m_s\frac{\vec{v}^2}{2}$.
We use the conventions from the *Intro to Plasma Physics* course from EPFL (Fasoli et al. 2018). An analogous outline of the derivation of the species equation can be found in chapter 2 of Freidberg 2014.

$$\Rightarrow \quad \frac{3}{2}n_s\frac{dT_s}{dt} + \overleftrightarrow{P}_s : \frac{\partial}{\partial \vec{r}}\vec{u}_s + \frac{\partial}{\partial \vec{r}}\cdot\vec{q}_s = Q_s\,,$$

with

- double contraction ":" : $\overleftrightarrow{P}_s : \frac{\partial}{\partial \vec{r}}\vec{u}_s := \sum_{i,j} P_{(s),\,ij}\frac{\partial u_{(s),\,i}}{\partial r_j}$

- $T_s := \frac{1}{3n_s}\int d\vec{v}\, m_s\,(\vec{v}-\vec{u}_s)^2 f_s$, the kinetic energy due to spreading around fluid velocity
- $\vec{q}_s = \frac{1}{2}m_s \int d\vec{v}\,(\vec{v}-\vec{u}_s)^2(\vec{v}-\vec{u}_s)f_s$, the heat flux vector, so the kinetic energy that is transported with velocity of the particles
- $Q_s = \int d\vec{v}\,\frac{1}{2}m_s\,(\vec{v}-\vec{u}_s)^2 \left(\frac{\partial f}{\partial t}\right)_{coll}$, heat generated due to viscous forces

2.4. Single Fluid Model

2.4.1. Single Fluid Variables

- Mass density: $\varrho(\vec{r},t) = \sum_s n_s m_s$
- Center of Mass Velocity: $\vec{V}(\vec{r},t) = \frac{\sum m_s n_s \vec{u}_s}{\varrho}$
- Total electric charge density: $\rho = \sum_s n_s q_s$
- Total electric current density: $\vec{J} = \sum_s q_s n_s \vec{u}_s$
- Pressure tensor:
  - Species pressure tensor in center of mass frame:
  $$\overleftrightarrow{P}_s^{cm} = \int d\vec{v}\, m_s\,(\vec{v}-\vec{V})\otimes(\vec{v}-\vec{V})f_s(\vec{r},\vec{v},t)$$
  - Singe fluid pressure tensor in center of mass frame: $\overleftrightarrow{P}^{cm} = \sum_s \overleftrightarrow{P}_s^{cm}$

2.4.2. Single Fluid Equations

In the following we want to combine the species equations to a model describing the behaviour of the combined fluid consisting of the different species.

Mass Continuity Equation

Multiplying the species continuity equation by their respective mass m_s, which is time independent, and adding the the equations for all present species, we obtain the *single fluid mass continuity equation*

$$\frac{\partial \varrho}{\partial t} + \nabla \cdot (\varrho \vec{V}) = 0\,.$$

Charge Continuity Equation

Similarly, multiplying the species continuity equation by their respective charges q_s, which is time independent, and adding the the equations for all present species, we obtain the *single fluid charge continuity equation*

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0\,.$$

Momentum Equation

Summing the species momentum equations (2.6), we can obtain the *single fluid momentum equation.*

$$\varrho \frac{d\vec{V}}{dt} = \rho \vec{E} + \vec{J} \times \vec{B} - \partial_{\vec{r}} \cdot \overleftrightarrow{P}^{cm} . \tag{2.7}$$

Details of the derivation can be found in A.2.

Ohm's Law

From the species momentum equations (2.6), assuming we have electrons and only one species of ions, we can multiply the electron momentum equation by $\frac{1}{q_e}$ and the ion momentum equation by $\frac{1}{q_i} = \frac{m_e}{m_i n_e}$. Further assuming neutrality ($n_e \simeq n_i \simeq n$), $m_e \ll m_i$, $\vec{J} \ll e n_e \vec{u}_e$, $e n_i \vec{u}_i$, the sum of the intermediate results is approximately

$$\frac{m_e}{ne^2} \left(\frac{\partial \vec{J}}{\partial t} + \frac{\partial}{\partial \vec{r}} \cdot (\vec{V}\vec{J} + \vec{J}\vec{V}) \right) = \vec{E} + \vec{V} \times \vec{B} - \frac{1}{en} \frac{\partial}{\partial \vec{r}} \cdot \overleftrightarrow{P}_e^{cm} - \eta \vec{J} ,$$

with the *plasma resisitivity* η. A more detailed explanation can be found in chapter 2 of Freidberg 2014.

Closure

In order to have a closed system of equations, we need additional assumptions and thus restrictions. This is called a *closure*. Many types of closures exist and most are quite involved.
For an isotropic pressure tensor and adiabatic processes we have the following closure:

$$\frac{\partial}{\partial \vec{r}} \cdot \overleftrightarrow{P}^{CM} = \frac{\partial}{\partial \vec{r}} P \quad , \quad \frac{d}{dt}(P \varrho^{-\gamma}) = 0 \quad \text{with} \quad \gamma = \frac{c_p}{c_v} = \frac{5}{3} .$$

We will justify these assumptions in the next section.

Single Fluid Maxwell Equations

For completeness, we also state the microscopic Maxwell equations again with total electric current density $\vec{J}$:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} , \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} , \tag{2.8}$$

$$\nabla \cdot \vec{B} = 0 , \qquad \nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) . \tag{2.9}$$

2.5. Magnetohydrodynamic Model

2.5.1. Magnetohydrodynamic Assumptions

By comparing constants as well as time and length scales typical in fusion plasmas or provided specifically for polomacs by Elio et al. 2024, we can make a number of assumptions following chapter 2 of Freidberg 2014, and for quasi-neutrality the plasma physics introductory course by Fasoli et al. 2018.
For each assumption we state the physical condition and give its general justification; quantitative verification for the Polomac of assumptions that depend on later introduced concepts is deferred to Chapter 5, where the relevant dimensionless parameters are introduced and computed.
We use the following values:

- Characteristic length of the Polomac: $L \sim 0.04\,\mathrm{m}$ to $0.3\,\mathrm{m}$
  - These values correspond to diameters of the central plasma cylinder and the outer plasma regions, respectively(Elio et al. 2024).
  - If not otherwise specified, we will in each case use the more limiting value of the two extremes.
- We assume approximate thermal equilibrium: $T_e \approx T_i \approx 100\,\mathrm{eV}$
- We use the proton mass for m_i. This is consistent with the prototype of Deutelio intending to use Hydrogen for demonstration purposes (Elio et al. 2024). Values that scale with $1/\sqrt{m_i}$ will scale with approximately $1/\sqrt{2}$ for Deuterium instead of Hydrogen.

1. **Electron inertia negligible** ($m_e \to 0$)

   Electrons respond nearly instantaneously to field changes because their cyclotron period $\tau_{ce} = 2\pi/\omega_{ce}$ is far shorter than the characteristic MHD timescale $\tau_{\mathrm{MHD}} \sim L/v_{th,i}$, where $v_{th,i} = \sqrt{2k_B T_i/m_i}$ is the ion thermal velocity (derived from the exponent of a Maxwellian distribution) and L is the plasma length scale. The condition is

   $$\frac{f_{\mathrm{MHD}}}{f_{ce}} = \frac{v_{th,i}/L}{f_{ce}} \ll 1.$$

   For the Polomac the ion thermal velocity is

   $$v_{th,i} = \sqrt{\frac{2 \cdot 100\,\mathrm{eV} \cdot 1.6 \times 10^{-19}\,\mathrm{J\,eV^{-1}}}{1.67 \times 10^{-27}\,\mathrm{kg}}} \approx 1.4 \times 10^{5}\,\mathrm{m\,s^{-1}}.$$

   Using the more limiting, upper bound of the ECR heating frequency from Elio et al. 2024, which is $f_{ce} = 8\,\mathrm{GHz}$, and $L = 0.3\,\mathrm{m}$:

   $$\frac{f_{\mathrm{MHD}}}{f_{ce}} = \frac{v_{th,i}/L}{f_{ce}} \approx 6 \times 10^{-5} \ll 1.$$

Ion dynamics are slow compared to the electron cyclotron motion; electron inertia can be neglected on MHD timescales.

2. **Quasi-neutrality**

   We assume the plasma to be quasi-neutral, i.e. that the typical plasma length scale L is much greater than the Debye screening distance λ_D (see (2.5)), and outside of that screening distance $\frac{\rho}{ne} = \frac{n_e - n_i}{n_i} \ll 1$.

   To motivate how the condition $L \gg \lambda_D$ guarantees also the latter, note that the electron density responds to the electrostatic potential of a fixed background ion density n_i as $n_e = n_i \exp(e\Phi/k_B T_e) \approx n_i(1 + e\Phi/k_B T_e)$ (Fasoli et al. 2018). For this linearization to hold we had to assume that there are many particles inside a Debye sphere for the collective shielding to be statistically meaningful, $N_D = n \cdot \frac{4\pi}{3}\lambda_D^3 \gg 1$, since Fasoli et al. 2018 show $e\Phi/k_B T_e \sim 1/(4\pi N_D^{2/3}) \ll 1$. Hence $(n_e - n_i)/n_i \approx e\Phi/k_B T_e$.

   The screened (Yukawa) potential around any test charge takes the form $\Phi(r) \propto r^{-1} \exp(-r/\lambda_D)$ (Fasoli et al. 2018), so for $r \gg \lambda_D$ the exponential suppresses Φ and therefore $e\Phi/k_B T_e \to 0$. If the plasma length scale L satisfies $L \gg \lambda_D$, charge separation is exponentially suppressed throughout the bulk.

   We conclude that quasi-neutrality requires

   $$\frac{L}{\lambda_D} \gg 1, \qquad N_D = n \cdot \tfrac{4\pi}{3}\lambda_D^3 \gg 1.$$

   For the Polomac (Elio et al. 2024), with $n = 10^{20}\,\mathrm{m}^{-3}$, $k_B T_e = 100\,\mathrm{eV}$ and the approxiamtive formula below for λ_D given by Chen 2016:

   - $\lambda_D = 7430\,\mathrm{m}\,\sqrt{T_{\mathrm{eV}}/n} = 7430\sqrt{100/10^{20}} = 7.4 \times 10^{-6}\,\mathrm{m}$
   - $L/\lambda_D \geq 0.04\,\mathrm{m}/7.4 \times 10^{-6}\,\mathrm{m} \approx 5.4 \times 10^3 \gg 1$
   - $N_D = 10^{20}\,\mathrm{m}^{-3} \cdot \frac{4\pi}{3}(7.4 \times 10^{-6}\,\mathrm{m})^3 \approx 1.7 \times 10^5 \gg 1$

   Both conditions are well satisfied.

3. ***Ideal* MHD - zero resistivity** ($\eta = 0$)

   Resistive diffusion is negligible when the magnetic field moves instantly with the plasma, i.e. when the resistive diffusion timescale greatly exceeds the Alfvén timescale. This is valid when the Lundquist number $S \gg 1$, the calculation of which is deffered to section 5.2.

   We can however already calculate the resistivity. The classical Spitzer resistivity (Chen 2016) is

   $$\eta_{\parallel} = 5.2 \times 10^{-5}\,\frac{Z \ln \Lambda}{T_{eV}^{3/2}} \quad [\Omega\text{-m}],$$

   where $\ln \Lambda \equiv \ln(r_{\max}/r_{\min})$ is the Coulomb logarithm, the ratio of the maximum to minimum effective impact parameters in a Coulomb collision (Beresnyak 2023).

For electron–ion collisions in the regime $T_i m_e/m_i < 10Z^2\,\mathrm{eV} < T_e$, which applies here since $T_i m_e/m_i \approx 0.05\,\mathrm{eV}$ and $T_e = 100\,\mathrm{eV} \gg 10\,\mathrm{eV}$ (Beresnyak 2023):

$$\ln\Lambda_{ei} = 24 - \ln\left(n_e^{1/2}\,T_e^{-1}\right) = 24 - \ln\left(\frac{\sqrt{10\times 10^{14}\,\mathrm{cm}^{-3}}}{100}\right) \approx 12.5.$$

For the Polomac ($Z = 1$, $T_e = 100\,\mathrm{eV}$):

$$\eta_\parallel = 5.2\times 10^{-5}\cdot\frac{1\cdot 12.5}{(100)^{3/2}} \approx 6.5\times 10^{-7}\,\Omega\,\mathrm{m},$$

and since $\eta_\perp \sim 2\eta_\parallel$ Chen 2016 we use $\eta \sim 10^{-6}\,\Omega\,\mathrm{m}$. Whether this justifies $\eta \to 0$ is determined by S; see Section 5.2.

4. **Neglect displacement current**

   In Ampère's law (2.9), $\nabla\times\vec{B} = \mu_0\vec{J} + \mu_0\epsilon_0\,\partial_t\vec{E}$, we drop the displacement current when $\vec{J}$ dominates. To estimate E, we use Faraday's law $\nabla\times\vec{E} = -\partial_t\vec{B}$, which gives $E/L \sim B/\tau \sim vB/L$, hence $E \sim vB$. Then with $c^2 = 1/\epsilon_0\mu_0$:

   $$\frac{\mu_0\epsilon_0\,\partial_t\vec{E}}{\nabla\times\vec{B}} \sim \frac{v^2B/c^2L}{B/L} = \frac{v^2}{c^2}.$$

   The fastest characteristic speed in MHD is the Alfvén velocity $v_A = \frac{B}{\sqrt{\mu_0\varrho}}$ (Freidberg 2014; Chen 2016), which is derived in section 5.1, so the condition reduces to $v_A^2/c^2 \ll 1$.

   Physically, dropping the displacement current filters out electromagnetic waves propagating at c, retaining only the slow MHD dynamics. The quantitative verification for the Polomac is also given in section 5.1.2.

5. **Small Larmor radius** ($L/\rho_i \gg 1$)

   The particles are tightly bound to the magnetic field lines. Inserting the definition of the larmor radius ($r_i = m_i v_{th,i}/eB$) this leads to

   $$\left|\vec{V}\times\vec{B}\right| \gg \left|\frac{1}{en}\frac{\partial}{\partial\vec{r}}\cdot\overleftrightarrow{P_e}\right|$$

   in Ohm's law. We see this by recognizing the larmor radius in

   $$\frac{\vec{V}\times\vec{B}}{\frac{1}{en}\frac{\partial}{\partial\vec{r}}\cdot\overleftrightarrow{P_e}} \sim \frac{v_{th,i}B}{\frac{n_e k_B T_e}{enL}} \sim \frac{v_{th,i}B}{\frac{k_B T_e}{eL}} \sim \frac{L}{r_i},$$

   where we used the ideal gas relation to estimate $\overleftrightarrow{P_e}$, the relation $T_e \sim T_i$ and $k_B T_i \sim m_i v_{th,i}^2$.

   With $L = 0.04\,\mathrm{m}$ to $0.3\,\mathrm{m}$ and $B = 0.2\,\mathrm{T}$ to $0.3\,\mathrm{T}$ this gives $L/r_i \approx 5.5$ to 62. The condition is satisfied at device scales; at the smallest resolved scales the margin is reduced and the ideal MHD description should be treated with appropriate caution.

6. **Isotropic pressure and incompressible motion**

   Isotropic pressure: The ideal MHD model assumes that the pressure tensor is isotropic, $\overleftrightarrow{P}^{CM} = P$, where P is a scalar pressure. This assumption is not rigorously justified for low-collisionality fusion plasmas, where pressure can become anisotropic ($P_\parallel \neq P_\perp$). However, as shown in Freidberg 2014 Section 2.4.3, MHD stability predictions remain qualitatively accurate because the most unstable modes couple to incompressible motions where the shortcomings of the isotropic pressure assumption have minimal impact.

   Adiabatic closure: From the ideal gas law, we see that the adiabatic equation of state ($\frac{ds}{dt} = 0$) is equivalent to $\frac{d}{dt}(\frac{P}{\varrho^\gamma})$ (see section A.3).

   and mass conservation, we obtain

$$\frac{d}{dt}\left(\frac{P}{\varrho^\gamma}\right) = 0 \iff \frac{dP}{dt} + \gamma P \nabla \cdot \vec{V} = 0\,.$$

   Because particles can move freely along $\vec{B}$ but are bound to field lines by gyromotion, a more physically accurate model would assume rapid heat flow along magnetic field lines ($\vec{B} \cdot \nabla T = 0$) rather than adiabatic behavior.

   Incompressible approximation: For incompressible motions ($\nabla \cdot \vec{V} = 0$) and perfect conductivity ($\eta = 0$), both the adiabatic closure and the infinite parallel thermal conductivity model reduce to the same result (see Freidberg 2014, Section 2.4.4):

$$\frac{dP}{dt} = 0\,.$$

   An analysis in Chapter 8.9 of Freidberg 2014 shows that geometries with closed field lines (such as the Polomac) support both compressible and incompressible instabilities, but the most unstable modes are typically incompressible. For symmetry-preserving modes that cannot be fully incompressible, quantitative stability predictions using ideal MHD should be viewed with caution, though qualitative trends remain valid. Because the isothermal equation of state (from ideal gas law) is $\frac{d}{dt}(\frac{P}{\varrho})$, so similar but with $\gamma = 1$, we conclude that whenever a result depends on γ, isothermal and adiabatic assumptions produce different outputs and we have to be careful with the result.

   In static equilibrium ($\vec{V} = 0$), the incompressibility condition is automatically satisfied.

   If more accurate models are required for the fusion-relevant regime, kinetic MHD or double adiabatic (CGL) theory can be used. Details can be found in Freidberg 2014.

2.5.2. Simplifications

- Charge continuity equation:

Charge continuity reduces to $\nabla \cdot \vec{J} = 0$, which is automatically satisfied by Ampere's law (Ass. 4: $\nabla \cdot (\nabla \times \vec{B}) = 0$). It is thus obsolete.

$$\underbrace{\frac{\partial \rho}{\partial t}}_{\text{Ass. 2}} + \underbrace{\frac{\partial}{\partial \vec{r}} \cdot \vec{J} = 0}_{\text{Ass. 4}}$$

- Momentum equation:

$$\varrho \frac{d\vec{V}}{dt} = \underbrace{\rho \vec{E}}_{\text{Ass. 2}} + \vec{J} \times \vec{B} - \frac{\partial}{\partial \vec{r}} \cdot \underbrace{\overleftrightarrow{P}^{CM}}_{\text{Ass. 6}}$$

- Ohm's law:

$$\underbrace{\frac{m_e}{ne^2}\left[\frac{\partial \vec{J}}{\partial t} + \frac{\partial}{\partial \vec{r}} \cdot (\vec{V}\vec{J} + \vec{J}\vec{V})\right]}_{\text{Ass. 1}} = \vec{E} + \vec{V} \times \vec{B} - \underbrace{\frac{1}{en}\frac{\partial}{\partial \vec{r}} \cdot \overleftrightarrow{P_e^{CM}}}_{\text{Ass. 5}} - \underbrace{\eta \vec{J}}_{\text{Ass. 3}}$$

- Maxwell equations

$$\underbrace{\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}}_{\text{redundant}} \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0 \qquad \nabla \times \vec{B} = \mu_0 \left(\vec{J} + \underbrace{\epsilon_0 \frac{\partial \vec{E}}{\partial t}}_{\text{Ass. 4}} \right)$$

2.5.3. The Ideal MHD Equations

The ideal magnetohydrodynamic equations consist of

- the mass-continuity equation:

$$\frac{\partial \varrho}{\partial t} + \nabla \cdot (\varrho \vec{V}) = 0$$

- the momentum equation:

$$\varrho \frac{d\vec{V}}{dt} = \vec{J} \times \vec{B} - \nabla P \tag{2.10}$$

- Ohm's law:

$$\vec{E} + \vec{V} \times \vec{B} = 0 \tag{2.11}$$

- Closure equation:

$$\frac{d}{dt}\left(P\varrho^{-\gamma}\right) = 0$$

- Maxwell equations:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \tag{2.12}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} \tag{2.13}$$

$$\nabla \cdot \vec{B} = 0$$

3. Polomac Geometry and Specialization to Poloidal Configurations

3.1. Polomac Geometry

The magnetic field is decomposed as

$$\vec{B} = \vec{B}_{\text{pol}}(r, \varphi, z) + \vec{B}_{\text{tor}}(r, \varphi, z), \tag{3.1}$$

where $\vec{B}_{\text{pol}} = B_r(r, \varphi, z)\hat{e}_r + B_z(r, \varphi, z)\hat{e}_z$ denotes the poloidal component and $\vec{B}_{\text{tor}} = B_\varphi(r, \varphi, z)\hat{e}_\varphi$ denotes the toroidal component, directed along the azimuthal direction.

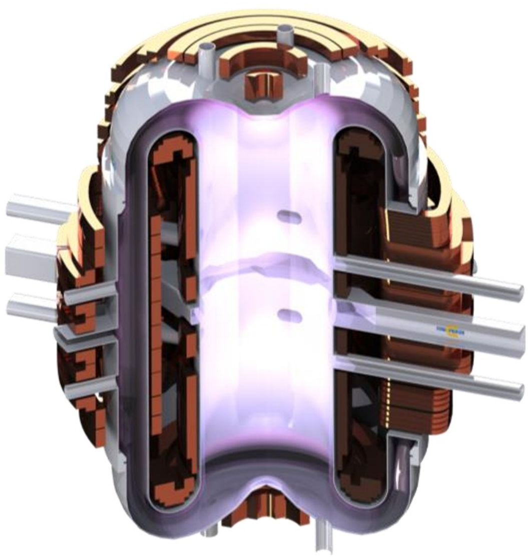

Figure 3.1.: Cut out view of the Polomac prototype, figure taken from Elio et al. 2024

From Figure 3.1 provided by Elio et al., we can see that the magnetic configuration of the Polomac is predominantly poloidal, so for $\vec{B} = B_r(r, \varphi, z)\hat{e}_r + B_\varphi(r, \varphi, z)\hat{e}_\varphi + B_z(r, \varphi, z)\hat{e}_z$, we have that in the middle sections $B_z \gg B_r, B_\varphi$ and in the top and bottom arcs $B_r \gg B_z, B_\varphi$. The total axisymmetry is broken in the middle section by the magnetic tunnels necessary to feed, support and cool the central solenoid. The device is described in greater detail in Elio et al. 2024. The arcs are axisymmetric, i.e., invariant under rotation about the z-axis.

3.2. Specialization to Poloidal Configurations

Since $B_\varphi \ll |\vec{B}_{\mathrm{pol}}|$, all terms in the MHD equations arising from the toroidal field component enter as corrections of order $\varepsilon \equiv B_\varphi / |\vec{B}_{\mathrm{pol}}| \ll 1$. To leading order, the dynamics are governed by the poloidal field alone. This is an important simplification and is used in all parts of chapter 4.

4. Static Equilibrium Model in Cylindrical Coordinates under Axisymmetric Assumptions

4.1. Axisymmetric and Poloidal Formulation

In the following we will assume the magnetic configuration to be axisymmetric, i.e. independent of the toroidal angle φ.
The magnetic field thus simplifies to

$$\vec{B} = B_r(r,z)\hat{e}_r + B_z(r,z)\hat{e}_z + B_\varphi(r,z)\hat{e}_\varphi \, . \tag{4.1}$$

.
In cylindrical coordinates (r, φ, z), the gradient, divergence and curl are:

$$\begin{aligned}
\nabla f &= \frac{\partial f}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial f}{\partial \varphi}\hat{e}_\varphi + \frac{\partial f}{\partial z}\hat{e}_z \, , \\
\nabla \cdot \vec{F} &= \frac{1}{r}\frac{\partial (rF_r)}{\partial r} + \frac{1}{r}\frac{\partial F_\varphi}{\partial \varphi} + \frac{\partial F_z}{\partial z} \, , \\
\nabla \times \vec{F} &= \left(\frac{1}{r}\frac{\partial F_z}{\partial \varphi} - \frac{\partial F_\varphi}{\partial z}\right)\hat{e}_r + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}\right)\hat{e}_\varphi + \frac{1}{r}\left(\frac{\partial (rF_\varphi)}{\partial r} - \frac{\partial F_r}{\partial \varphi}\right)\hat{e}_z \, ,
\end{aligned}$$

so

$$\nabla \cdot \vec{B} = \frac{1}{r}\frac{\partial (rB_r)}{\partial r} + \frac{\partial B_z}{\partial z} \, , \tag{4.2}$$

$$\nabla \times \vec{B} = \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right)\hat{e}_\varphi - \frac{\partial B_\varphi}{\partial z}\,\hat{e}_r + \frac{1}{r}\frac{\partial (rB_\varphi)}{\partial r}\,\hat{e}_z \, . \tag{4.3}$$

In our axisymmetric approximation we disregard any impact the tunnels have. Since the polomac does not include any toroidal field coils, the only possible source of a nonzero $B_\varphi(r,z)\hat{e}_\varphi$ are these tunnels, which we are disregarding (so zeroth order in our poloidal expansion).
We thus set $B_\varphi(r,z)\hat{e}_\varphi \equiv 0$, which further simplifies (4.1) to

$$\vec{B} = B_r(r,z)\hat{e}_r + B_z(r,z)\hat{e}_z \, , \tag{4.4}$$

and (4.3) to

$$\nabla \times \vec{B} = \left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right)\hat{e}_\varphi \, . \tag{4.5}$$

4.2. Static Equilibrium

For the following derivation we closely work with Chapter 6 of Freidberg 2014 which derives the Grad-Shafranov equation. Due to our special geometry, we can make a couple of simplifications.
In case of a static equilibrium ($\vec{V} = 0$, $\frac{\partial}{\partial t}\cdot = 0$), of the ideal MHD equations only the following *static equilibrium equations* remain.

- the static momentum equation:

$$\vec{J} \times \vec{B} = \nabla P \tag{4.6}$$

- static Maxwell equations:
  –
  $$\nabla \times \vec{B} = \mu_0 \vec{J} \tag{4.7}$$
  –
  $$\nabla \cdot \vec{B} = 0 \tag{4.8}$$

Using (4.2), (4.4) and (4.5) we can rewrite these equations and only the following components remain:

- $\nabla \cdot \vec{B} = 0$ becomes:

$$\frac{1}{r}\frac{\partial (rB_r)}{\partial r} + \frac{\partial B_z}{\partial z} = 0 \tag{4.9}$$

- Ampère's law $\nabla \times \vec{B} = \mu_0 \vec{J}$ gives a purely toroidal current density:

$$\vec{J} = J_\varphi(r,z)\,\hat{e}_\varphi, \qquad J_\varphi = \frac{1}{\mu_0}\left(\frac{\partial B_r}{\partial z} - \frac{\partial B_z}{\partial r}\right) \tag{4.10}$$

- The momentum equation $\vec{J} \times \vec{B} = \nabla P$ yields

$$J_\varphi\,\hat{e}_\varphi \times (B_r\,\hat{e}_r + B_z\,\hat{e}_z) = \frac{\partial P}{\partial r}\hat{e}_r + \frac{\partial P}{\partial z}\hat{e}_z\,. \tag{4.11}$$

  Using $\hat{e}_\varphi \times \hat{e}_r = -\hat{e}_z$ and $\hat{e}_\varphi \times \hat{e}_z = \hat{e}_r$, this gives two scalar equations:

$$J_\varphi B_z = \frac{\partial P}{\partial r} \tag{4.12}$$
$$-J_\varphi B_r = \frac{\partial P}{\partial z} \tag{4.13}$$

Notice that (4.9) implies that $\vec{B}$ can be written identically by introducing the poloidal flux function $\psi(r,z)$ such that

$$\vec{B} = \frac{1}{r}\nabla\psi \times \hat{e}_\varphi, \qquad B_r = -\frac{1}{r}\frac{\partial \psi}{\partial z}, \quad B_z = \frac{1}{r}\frac{\partial \psi}{\partial r}. \tag{4.14}$$

By construction, $\vec{B} \cdot \nabla\psi = 0$, so ψ is constant along field lines.
Dotting $\vec{B}$ into the momentum equation (4.6) gives $\vec{B} \cdot \nabla P = 0$, so P is constant along field lines as well.
We assume that in the plasma domain the field lines form closed, nested loops around the torus of central coils, consistent with a dipole-like topology, but not derived from ideal MHD. Under this assumption each field line loop has a unique value of ψ, making ψ a valid flux surface label and $P(\psi)$ a well-defined single-valued function.
The function $P(\psi)$ cannot be determined from ideal MHD alone. As Freidberg 2014 notes, transport theory, experimental data, or physical intuition are needed to specify it.

Remark If we did not assume $B_\varphi \equiv 0$, we would have restricted the above discussion to B_{pol} and added a similar function for B_φ (see Freidberg 2014). The argument for nested flux surfaces would get more involved.

4.2.1. Reduced Grad-Shafranov Equation

We can rewrite (4.10) as

$$\mu_0 J_\varphi = -\frac{1}{r}\Delta^*\psi\,, \tag{4.15}$$

where, using cylindrical divergence again,

$$\Delta^*\psi := r^2\nabla \cdot \left(\frac{\nabla\psi}{r^2}\right) = r\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial\psi}{\partial r}\right) + \frac{\partial^2\psi}{\partial z^2}\,.$$

If we furthermore use the chain rule ($\nabla P = \frac{dP}{d\psi}\nabla\psi$) and (4.15), we see that for non-trivial $\psi(r,z)$ both components of the momentum equation ((4.12), (4.13)) are satisfied simultaneously if and only if

$$\Delta^*\psi = -\mu_0 r^2\frac{dP}{d\psi}. \tag{4.16}$$

We call this the *reduced Grad-Shafranov equation.*

Remark Instead of specifying an experimentally obtained pressure profile $P(\psi)$, we could also specify an experimentally obtained current profile $J_\varphi(r,z)$, insert it in (4.15) and recover the pressure via (4.14) and the static momentum equation (4.6). This does not change the fact, that MHD is underdetermined.

4.2.2. Boundary Conditions

Geometric Axis ($r = 0$)

For $\vec{B}$ and p to remain finite on the symmetry axis, we require

$$\nabla\psi \xrightarrow{r\to 0} 0.$$

This is a regularity condition, not a physical wall.

Conducting Walls (Inner Coil Surface and Outer Vacuum Vessel)

Assuming the wall materials are perfect conductors and initially there is no magnetic field inside the wall, the integral version of Gauss' law for magnetism (2.2) gives

$$\vec{B} \cdot \hat{n} = 0 \quad \text{on all walls} \tag{4.17}$$

In the static equilibrium ($\partial_t B = 0$) this holds for all times.
In our axisymmetric setup, the wall normal $\hat{n} = n_r \hat{e}_r + n_z \hat{e}_z$ has no φ-component. We can recognize the wall tangent in the poloidal plane $\hat{t} = \hat{e}_\varphi \times \hat{n}$ when we insert (4.14) into (4.17) and use the triple product

$$0 = (\nabla\psi \times \hat{e}_\varphi) \cdot \hat{n} = \hat{t} \cdot \nabla\psi \,,$$

and thus ψ is constant along the walls in the poloidal plane. Because ψ is only a function of r, z, it is thus constant on the entire inner and outer wall, respectively.

5. Relevant Parameters for the Polomac

For the following three dimensionless (or characteristic) parameters governing the MHD dynamics of the Polomac we first introduce the underlying physics, then evaluate the parameter using design values from Elio et al. 2024, and finally state which assumption of subsection 2.5.1 it justifies.
Throughout this chapter we use the following Polomac values (Elio et al. 2024):

- Hydrogen plasma: $m_i = 1.67 \times 10^{-27}\,\mathrm{kg}$, $Z = 1$
- Plasma density: $n = 1 \times 10^{20}\,\mathrm{m}^{-3}$
- Temperature: $T_i \approx T_e = 100\,\mathrm{eV}$
- Magnetic field: $B = 0.2\,\mathrm{T}$ to $0.3\,\mathrm{T}$
- Characteristic length: $L \sim 0.04\,\mathrm{m}$ to $0.3\,\mathrm{m}$

Note that some of these values $(n, T, (B))$ are target values and not yet experimentally verified.

5.1. Alfvén Velocity and Waves

5.1.1. Theory of Alfvén waves

In this subsection we follow chapter 8 of Freidberg 2014.
Alfvén waves are transverse, incompressible oscillations that propagate along magnetic field lines, driven by magnetic tension and resisted by plasma inertia. To derive their dispersion relation we linearize the ideal MHD equations around a static uniform equilibrium $\vec{B} = B_0\hat{e}_z$, $\varrho = \varrho_0$, $\vec{V} = 0$, $P = P_0$, writing each quantity as $Q = Q_0 + Q_1$, so the the equilibrium value plus a small perturbation (e.g. $\vec{B} = B_0\hat{e}_z + \vec{B}_1$). To first order, combining the ideal MHD Ohm's law (2.11) and Faraday's law (2.12), we get the so called induction equation

$$\frac{\partial \vec{B}_1}{\partial t} = \nabla \times (\vec{V}_1 \times \vec{B}_0), \tag{5.1}$$

and the ideal MHD momentum equation (2.10) combined with the ideal MHD Ampère-Maxwell law (2.13) gives the

$$\varrho_0 \frac{\partial \vec{V}_1}{\partial t} = -\nabla P_1 + \frac{1}{\mu_0}(\nabla \times \vec{B}_1) \times \vec{B}_0. \tag{5.2}$$

We seek plane-wave solutions $Q_1 \propto e^{i(\vec{k}\cdot\vec{x}-\omega t)}$, thus without loss of generality placing $\vec{k} = k_\perp \hat{e}_y + k_\parallel \hat{e}_z$. The shear Alfvén branch is polarised perpendicular to both $\vec{B}_0$ and $\vec{k}$, i.e. $\vec{V}_1 = V_1 \hat{e}_x$. Because $\vec{k}\cdot\vec{V}_1 = 0$ the motion is incompressible and $P_1 = 0$, so the pressure gradient in the momentum equation (5.2) drops out.
Substituting the plane-wave ansatz into a fourier transformed induction equation ($\partial_t \to -i\omega$, $\nabla \to i\vec{k}$) gives

$$\vec{B}_1 = -\frac{1}{\omega}\,\vec{k}\times(\vec{V}_1\times\vec{B}_0).$$

With $\vec{V}_1 = V_1\hat{e}_x$ and $\vec{B}_0 = B_0\hat{e}_z$:

$$\vec{V}_1\times\vec{B}_0 = -V_1 B_0\,\hat{e}_y \quad\Longrightarrow\quad \vec{B}_1 = -\frac{k_\parallel B_0}{\omega}\,V_1\,\hat{e}_x.$$

Inserting $\vec{B}_1$ into the $\hat{e}_x$-component of the fourier transformed momentum equation:

$$-i\omega\varrho_0 V_1 = \frac{i}{\mu_0}\big[(\vec{k}\times\vec{B}_1)\times\vec{B}_0\big]_x = \frac{i}{\mu_0}\,k_\parallel B_0\,B_{1x} = -\frac{ik_\parallel^2 B_0^2}{\mu_0\omega}\,V_1.$$

Dividing through by $-i\varrho_0 V_1/\omega$ yields the dispersion relation

$$\omega^2 = k_\parallel^2\,v_A^2, \tag{5.3}$$

where

$$v_A = \frac{B_0}{\sqrt{\mu_0\varrho_0}} \tag{5.4}$$

is the *Alfvén velocity.* The relation (5.3) is non-dispersive: phase and group velocities are equal and directed along $\vec{B}_0$. Only the parallel wavenumber $k_\parallel$ enters, so the wave is insensitive to $k_\perp$. The perturbation involves no compression, no density fluctuation, and no pressure fluctuation; it is a pure field-line-bending mode, oscillating between perpendicular kinetic energy and magnetic tension energy.

5.1.2. Alfvén Velocity of the Polomac

With $B = 0.25\,\mathrm{T}$ (midpoint of operating range) and $\varrho = nm_i = 10^{20}\,\mathrm{m}^{-3}\cdot 1.67\times 10^{-27}\,\mathrm{kg} = 1.67\times10^{-7}\,\mathrm{kg\,m}^{-3}$ we get for (5.4):

$$v_A \approx 5.5\times10^5\,\mathrm{m\,s}^{-1}.$$

The corresponding MHD timescale for $L = 0.3\,\mathrm{m}$ is $\tau_A = L/v_A \approx 5.5\times10^{-7}\,\mathrm{s}$.

Justification of Neglection of Displacement Current in MHD Assumption 4

$$\frac{v_A}{c} = \frac{5.5\times10^5}{3\times10^8} \approx 1.8\times10^{-3} \ll 1.$$

Thus the MHD dynamics are non-relativistic and the displacement current in Ampère's law is negligible.

5.2. Lundquist Number

Physics

The Lundquist number (see Goldston and Rutherford 1995) compares the resistive diffusion timescale $\tau_R = \mu_0 L^2/\eta$ (from $\frac{B}{\tau_R} \sim \frac{\eta}{\mu_0}\frac{B}{L^2}$) to the Alfvén timescale $\tau_A = L/v_A$:

$$S := \frac{\tau_R}{\tau_A} = \frac{\mu_0 L v_A}{\eta}. \tag{5.5}$$

Equivalently, S is the magnetic Reynolds number evaluated at the Alfvén speed. When $S \gg 1$, resistive diffusion is slow on MHD timescales and the magnetic flux is effectively frozen into the fluid, the ideal MHD limit.

5.2.1. Lundquist Number of the Polomac

Using $\eta \sim 1 \times 10^{-6}\,\Omega\,\mathrm{m}$ (Section 2.5.1), $v_A = 5.5 \times 10^5\,\mathrm{m\,s^{-1}}$, and $L = 0.04\,\mathrm{m}$ to $0.3\,\mathrm{m}$

$$S \approx 2.8 \times 10^4 \text{ to } 2.1 \times 10^5.$$

Justification of Zero resistivity in Ideal MHD Assumption

$S \in [2.8 \times 10^4, 2.1 \times 10^5]$ means the resistive diffusion timescale τ_R exceeds the Alfvén timescale τ_A by four to five orders of magnitude. Magnetic flux is therefore effectively frozen into the plasma on MHD timescales, justifying $\eta \to 0$.

5.3. Plasma β

5.3.1. Physics

The plasma β is the ratio of kinetic plasma pressure to magnetic pressure and measures confinement efficiency. A higher β means more fusion power per unit of magnetic field energy invested. There is no universal definition of plasma β, the definition found in Freidberg 2014 is for example tailored to tokamaks, where the dominant toroidal field sets the magnetic pressure scale. For the Polomac, which operates in a predominantly poloidal field with $B_\varphi \approx 0$, a natural definition is the local ratio

$$\beta(\vec{r}) := \frac{P(\vec{r})}{B(\vec{r})^2/(2\mu_0)}, \tag{5.6}$$

where P and B are evaluated at the same point. This is a *local* β, not a volume-averaged quantity. The distinction matters: while a global $\beta > 1$ would imply that the plasma pressure exceeds the total confining magnetic pressure and is therefore not achievable in steady-state magnetic confinement, a *local* $\beta > 1$ is physically admissible wherever the field has a near-null, and has been experimentally demonstrated in the RT-1 experiment Nishiura et al. 2015. Note also that comparing β values across geometries requires care, since the dominant field component (toroidal vs. poloidal) sets the denominator differently in each case (Freidberg 2014). However, the dominating fields usually also dominate (lower the total value) β (Freidberg 2014), so comparisons of efficiency still make sense.

5.3.2. Plasma β of the Polomac

Since the ideal Grad-Shafranov equation does not yield explicit values of P and B without a numerical solution, we estimate β using the prototype parameter ranges given in Elio et al. 2024 (Table 1): plasma pressure $P \in [1.6\,\mathrm{kPa},\ 16\,\mathrm{kPa}]$ and magnetic flux density $B \in [0.2\,\mathrm{T},\ 0.3\,\mathrm{T}]$.
Inserting into (5.6), the envelope of achievable β values is

$$\beta \in [0.04,\ 1.01],$$

where the lower bound corresponds to $P = 1.6\,\mathrm{kPa}$, $B = 0.3\,\mathrm{T}$ and the upper bound to $P = 16\,\mathrm{kPa}$, $B = 0.2\,\mathrm{T}$. These are not a single operating point but the extremes of the parameter envelope. The upper end of this range, $\beta \lesssim 1$, is consistent with the high-β regime characteristic of levitated-dipole configurations Nishiura et al. 2015; Garnier, Hansen, et al. 2006.

Significance for Future Fusion Devices

If the upper range $\beta \to 1$ is achievable, the Polomac could confine fusion-grade pressures with a comparatively modest magnetic field, reducing both engineering complexity and power consumption. These are design targets and have not yet been experimentally demonstrated.

6. An Overview over Magnetic Confinement Approaches for Nuclear Fusion

6.1. Relevant factors

6.1.1. Lawson Criterion

High performance, as quantified by the Lawson criterion, is a prerequisite for any viable fusion reactor. For the D–T reaction the ignition condition can be written as

$$P\tau_E \gtrsim 8 \times 10^5\,\mathrm{Pa\,s}, \tag{6.1}$$

where P is the plasma pressure and τ_E the energy confinement time Freidberg 2014. The function $T^2/\langle\sigma v\rangle$ that enters the right-hand-side in the full expression of (6.1) has a minimum at approximately $T \approx 15$ keV; operating near this temperature minimises the required $P\tau_E$ and therefore the cost of the device (Freidberg 2014). Achieving high P is the concern of MHD confinement, while achieving long τ_E is the concern of transport physics, and both must be satisfied simultaneously.
See section 5.3 for a discussion on plasma β comparison.

6.1.2. Steady-State Operation

Steady-state operation would reduce the cyclic thermal and mechanical stresses of components and, given it does not create other unforeseen troubles that are avoided in pulsed operations, is economically more viable (Freidberg 2014).
For configurations that rely on an inductively driven toroidal plasma current, such as the standard tokamak, the finite volt-second capacity of the central solenoid limits each discharge to a finite pulse duration. Configurations that do not rely on a driven toroidal plasma current, such as the stellarator, avoid this constraint and are intrinsically steady-state Freidberg 2014.

6.2. Toroidal Geometries

A toroidal setup, so basically an end-to-end connected Z-pinch, has many nice properties, which is why many current experiments follow this torus ansatz. However, in purely toroidal fields, due to asymmetries of the magnetic on the inside versus the outside of the torus, an outside directed $\vec{E} \times \vec{B}$-drift is created (see Freidberg 2014, section 4.9).

The tokamak and stellarator have different approaches to overcome this challenge and to create a flux of particles from top to bottom of the torus and vice versa, thus avoiding the voltage build-up along the vertical axis and thus the $\vec{E} \times \vec{B}$-drift.

6.2.1. Tokamak Ansatz

Confinement in a Tokamak

A tokamak magnetic field is the superposition of a strong externally-applied toroidal field B_{tor} (produced by toroidal-field coils external to the plasma) and a weaker poloidal field B_{pol} produced by the toroidal plasma current I_{tor} and controlled and shaped by external poloidal-field coils (Freidberg 2014).
The ratio is captured by the safety factor $q(r) \approx rB_{\text{tor}}/(RB_{\text{pol}})$ where r, R are the minor and major radius of the torus, respectively. A safety factor of at least $q = 1$ is needed to reduce instabilities of current-driven modes.
Flux surfaces are nested, toroidal surfaces and the magnetic field lines are helices on them. The safety factor describes how many times a line is winded around the central toroidal torus axis compared to around the symmetric axis located at $r = 0$.
A safety factor of $q \gtrsim 1$ is typically reached in the innermost flux surfaces, and it increases for flux surfaces further out. A more detailed discussion can be found in Chapter 5.5.2 of Freidberg 2014.

Equilibrium Formulation and Steady-State Operation

The equilibrium of an axisymmetric toroidal plasma is governed by the full Grad–Shafranov equation (Freidberg 2014):

$$\Delta^* \psi = -\mu_0 R^2 \frac{dp}{d\psi} - \frac{1}{2}\frac{dF^2}{d\psi}, \tag{6.2}$$

where $\psi(R, Z)$ is the poloidal flux function, $p(\psi)$ the plasma pressure, and $F(\psi) = RB_{\text{tor}}$ encodes the toroidal field. This second-order nonlinear PDE, derived from ideal MHD force balance $\vec{J} \times \vec{B} = \nabla p$ under axisymmetry, simultaneously describes radial pressure balance and toroidal force balance; its solutions give the nested flux surfaces on which p and F are constant (Freidberg 2014, Chapter 6).
As described in subsection 6.1.2, the standard tokamak is not intrinsically steady-state. Steady-state operation requires non-inductive current drive, either external (neutral-beam or radio-frequency) or via the self-generated bootstrap current, which arises from pressure-gradient-driven particle drifts in the neoclassical transport picture. The so-called *advanced tokamak* (AT) scenario exploits a large bootstrap fraction together with a reversed-shear $q(r)$ profile to approach steady-state operation (Freidberg 2014).

Stability Considerations and Disruption Risks

Tokamak stability is limited by both current-driven and pressure-driven instabilities.

On the current side, the safety factor q must remain above a minimum value throughout the plasma, in practice $q \gtrsim 2-3$ at the edge, to suppress current-driven kink instabilities (Freidberg 2014, section 12.11). If too much current flows, these instabilities cause a major disruption: a rapid quench of both plasma current and pressure on a millisecond timescale, which is one of the principal engineering risks of the tokamak Freidberg 2014. On the pressure side, there is an empirical upper bound on β, the ratio of plasma pressure to magnetic pressure. For a tokamak this limit scales as $\beta_{\max} \propto I_p/(aB_T)$, so higher current and weaker field permit higher pressure (Freidberg 2014). Exceeding this bound also leads to major disruptions.

Performance of Tokamaks

If successful, ITER, the next-step international experiment, is designed to achieve $Q = 10$ (fusion power ten times the external heating power), corresponding to $P\tau_E \approx 6.5 \times 10^5\,\mathrm{Pa\,s}$ (Freidberg 2014, subsection 6.6.6).
ITER targets a toroidal beta of $\beta_t \approx 0.025$, calculated via $\beta_t(\%) = \beta_N \cdot \frac{I_p}{aB_0}$ (Freidberg 2014) with the baseline normalized beta $\beta_N = 1.8$ (Sips et al. 2018).
The main performance limitation of the tokamak is the pulsed nature of inductive operation and the risk of major disruptions; both are active areas of research targeted by the AT and ITER programs (Freidberg 2014).

6.2.2. Stellarator Ansatz

Confinement in a Stellarator

Many different designs are grouped as Stellarators. What they have in common is that they do not try to induce a large toroidal currents to generate a poloidal electric field. Instead, they break the toroidal symmetry by reallocating of the coils inducing a toroidal field and that way are able to mix up vertical the distance of the particles from the center. They still have approximately nested flux surfaces (Freidberg 2014).
The stellarator analog of the safety factor is the *rotational transform* ι, defined via $q = 2\pi/\iota$. Unlike in a tokamak, B_{pol} is not constant along a field line in a stellarator, so the simple ratio $rB_{\text{tor}}/(RB_{\text{pol}})$ is not well-defined pointwise. Instead, one must take the field-line average over one helical period (Freidberg 2014):

$$\frac{\iota}{2\pi} = \frac{\displaystyle\oint \frac{B_{\text{pol}}}{rB}\,dl}{\displaystyle\oint \frac{B_{\text{tor}}}{RB}\,dl}. \tag{6.3}$$

A big engineering challenge is coil manufacture: the non-planar modular superconducting coils of W7-X, a current leading experiment, must be constructed to millimetre tolerance in order to control magnetic island widths and avoid confinement degradation.

Equilibrium Formulation and Steady-State Operation

Because the stellarator lacks toroidal symmetry, the Grad-Shafranov equation (6.2) no longer applies. The equilibrium is instead a full three-dimensional ideal-MHD force balance, $\vec{J} \times \vec{B} = \nabla p$, which must be solved in 3D.
Because stellarators carry no net driven toroidal current, they are intrinsically steady-state. A residual bootstrap current does arise from pressure-gradient-driven particle drifts, but in W7-X this has been minimised by design (Freidberg 2014). The practical consequence is that W7-X has sustained a plasma for 363 seconds with an energy throughput of 1.8 GJ in its OP 2.3 campaign (Grulke et al. 2026). During that campaign, volume averaged plasma β as approached 0.03 (Grulke et al. 2026).

Stability Considerations and Disruption Risks

Stellarators are subject to pressure-driven instabilities, interchange and ballooning modes, but are largely free of the current-driven kink instabilities that dominate tokamak stability physics, because they operate with zero or very small net toroidal current (Freidberg 2014).

6.3. Poloidal Geometries

Poloidal geometries feature elegant physics, but engineering challenges in regard to their central magnetic dipole. The Levitating Dipole Experiment (LDX) was terminated in 2011, but created possibly valuable results upon which new designs such as the Polomac or OpenStar's levitating dipole reactor (see Simpson et al. 2026) can build.

6.3.1. Polomac

Note that Elio et al. in Elio2024 are targeting much higher temperatures because they want to use D-D-fusion, for which the Lawson-criterion requires higher values (see Elio et al. 2024).
The confinement in the Polomac is achieved via the closed poloidal magnetic field lines, the MHD equilibrium has been discussed in chapter 4 and operation is in principle steady-state. The performance targets of the Polomac are outlined in Elio et al. 2024 and remain to be validated.

Stability Considerations and Disruption Risks

The Polomac, like the LDX, carries no driven toroidal plasma current. Current driven instabilities are thus less problematic.
The principal stability threat is instead the *interchange instability* (described in Rosenbluth and Longmire 1957). Because the magnetic field is purely poloidal and the rotational transform is identically zero, there is no magnetic shear and no average-good-curvature region available to suppress pressure-driven interchange modes (Freidberg 2014,

§ 8.11.3). The only stabilising mechanism is *plasma compressibility.* Stability requires the pressure profile to satisfy the adiabaticity condition

$$\frac{d}{d\psi}(p\,V^{\gamma}) \geq 0, \qquad \gamma = \tfrac{5}{3}, \quad V(\psi) = \oint \frac{dl}{B}, \tag{6.4}$$

where $V(\psi)$ is the specific volume of a flux tube (Rosenbluth and Longmire 1957; Garnier, Kesner, and Mauel 1999). Profiles steeper than $p \propto V^{-\gamma}$ are interchange-unstable; shallower profiles are stable. Further theoretical background on stability at this border can be found in Garnier, Kesner, and Mauel 1999, with physical confirmation for LDX provided in Garnier, Hansen, et al. 2006.

A second, Polomac-specific risk is the hot-electron interchange (HEI) instability. In the supported (non-levitated) operating mode of LDX, intense HEI bursts prevented the build-up of high-pressure plasma; they were suppressed only when the background plasma density was raised sufficiently (Garnier, Hansen, et al. 2006). The Polomac avoids plasma contact with the physical support structure using four magnetic tunnels, but these break toroidal axisymmetry in a way that could be qualitatively analogous to rod supports. Whether compressibility stabilisation survives this four-fold symmetry breaking has not yet been quantitatively assessed (Elio et al. 2024).

7. Framework for a Minimal FDTD Solver for Linearized MHD of the Polomac

7.1. Problem Formulation

We want to be able to solve for the independent time dependent properties of the MHD equations in subsection 2.5.3, which are ϱ, $\vec{V}$, $\vec{B}$ and P.
If we have $\vec{B}$ and $\vec{V}$ we will be able to solve for $\vec{E}$ and $\vec{J}$ at any time via (2.11) and (2.13), respectively.

7.1.1. Equations to be Solved

As in subsection 5.1.1 we eliminate $\vec{E}$ and $\vec{J}$. Additionally we take the mass continuity equation (2.5.3) and the closure equation (2.5.3) to arrive at a system of four partial differential equations

$$\begin{aligned}
\frac{\partial \varrho}{\partial t} + \nabla \cdot (\varrho \vec{V}) &= 0\,, \\
\varrho \frac{d\vec{V}}{dt} &= -\nabla P + \frac{1}{\mu_0}(\nabla \times \vec{B}) \times \vec{B}\,, \\
\frac{\partial \vec{B}}{\partial t} &= \nabla \times (\vec{V} \times \vec{B})\,, \\
\frac{d}{dt}\left(P\varrho^{-\gamma}\right) &= 0\,,
\end{aligned}$$

and as an additional constraint Gauss' law for magnetism (2.5.3), which is not evolved but must be preserved

$$\nabla \cdot \vec{B} = 0. \tag{7.1}$$

7.2. Linearization and Discretization

As in subsection 5.1.1, we linearize around a uniform static equilibrium

$$\begin{aligned}
\vec{B}(\vec{x},t) &= \vec{B}^{(0)} + \vec{B}^{(1)}(\vec{x},t),\\
\vec{V}(\vec{x},t) &= \vec{V}^{(1)}(\vec{x},t),\\
\varrho(\vec{x},t) &= \varrho^{(0)} + \varrho^{(1)}(\vec{x},t),\\
P(\vec{x},t) &= P^{(0)} + P^{(1)}(\vec{x},t),
\end{aligned}$$

where $\vec{B}^{(0)}$, $\varrho^{(0)}$, $P^{(0)}$ are constants, $\vec{V}^{(0)} = 0$, and, as in subsection 5.1.1, perturbations of all quantities $Q = Q_0 + Q_1$ satisfy satisfy $|Q_1/Q_0| \ll 1$. Substituting and retaining only first-order terms yields the linearized equations

$$\frac{\partial \varrho^{(1)}}{\partial t} = -\varrho^{(0)}\, \nabla \cdot \vec{V}^{(1)}, \tag{7.2}$$

$$\varrho^{(0)}\, \frac{\partial \vec{V}^{(1)}}{\partial t} = -\nabla P^{(1)} + \frac{1}{\mu_0}\big(\nabla \times \vec{B}^{(1)}\big) \times \vec{B}^{(0)}, \tag{7.3}$$

$$\frac{\partial \vec{B}^{(1)}}{\partial t} = \nabla \times \big(\vec{V}^{(1)} \times \vec{B}^{(0)}\big), \tag{7.4}$$

$$\frac{\partial P^{(1)}}{\partial t} = -\gamma P^{(0)}\, \nabla \cdot \vec{V}^{(1)}. \tag{7.5}$$

Equations (7.2) and (7.5) together imply $P^{(1)} = c_s^2 \varrho^{(1)}$, where $c_s = \sqrt{\gamma P^{(0)}/\varrho^{(0)}}$ is the adiabatic sound speed, so they carry the same information. The system is therefore closed by (7.3) and (7.4), with $P^{(1)}$ supplied by (7.5). The constraint $\nabla \cdot \vec{B}^{(1)} = 0$ is automatically preserved by (7.4) and serves as a numerical consistency check. The unknowns to be time-stepped are $\vec{V}^{(1)}$, $\vec{B}^{(1)}$, and $P^{(1)}$.

7.2.1. Reduction to 2D Axisymmetric Cylindrical Coordinates

The Polomac geometry is naturally described in cylindrical coordinates (r, φ, z). We adopt two simplifications consistent with the rest of this work.

First, we set $\partial/\partial\varphi = 0$, i.e. we restrict to axisymmetric perturbations. This is the same approximation already made in chapter 4 and is motivated by the axisymmetry of the Polomac, broken only by the tunnels (see chapter 3). All perturbation fields are then functions of (r, z, t) only.

Second, we choose $\vec{B}^{(0)} = B^{(0)}\hat{e}_z$, a uniform field along the z-axis. This is what is to be expected in the vertical parts of the Polomac (in the middle, outside the arcs) and is the simplest equilibrium consistent with a dominantly poloidal field. We used the same setting used in subsection 5.1.1 to derive the Alfvén dispersion relation.

Under these two assumptions the cylindrical differential operators from section 4.1 reduce to

$$\nabla \cdot \vec{F} = \frac{1}{r}\frac{\partial (rF_r)}{\partial r} + \frac{\partial F_z}{\partial z}, \tag{7.6}$$

$$(\nabla \times \vec{F})_r = -\frac{\partial F_\varphi}{\partial z}, \qquad (\nabla \times \vec{F})_\varphi = \frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}, \qquad (\nabla \times \vec{F})_z = \frac{1}{r}\frac{\partial (rF_\varphi)}{\partial r}. \tag{7.7}$$

We expand the perturbation fields as $\vec{V}^{(1)} = V_r\hat{e}_r + V_\varphi\hat{e}_\varphi + V_z\hat{e}_z$ and $\vec{B}^{(1)} = B_r^{(1)}\hat{e}_r + B_\varphi^{(1)}\hat{e}_\varphi + B_z^{(1)}\hat{e}_z$. Inserting $\vec{B}^{(0)} = B^{(0)}\hat{e}_z$ and evaluating (7.3)–(7.5) component by component using (7.6)–(7.7) yields the following seven scalar equations.

Momentum

$$\varrho^{(0)}\frac{\partial V_r}{\partial t} = -\frac{\partial P^{(1)}}{\partial r} + \frac{B^{(0)}}{\mu_0}\frac{\partial B_r^{(1)}}{\partial z} - \frac{B^{(0)}}{\mu_0}\frac{\partial B_z^{(1)}}{\partial r}, \tag{7.8}$$

$$\varrho^{(0)}\frac{\partial V_\varphi}{\partial t} = \frac{B^{(0)}}{\mu_0}\frac{\partial B_\varphi^{(1)}}{\partial z}, \tag{7.9}$$

$$\varrho^{(0)}\frac{\partial V_z}{\partial t} = -\frac{\partial P^{(1)}}{\partial z}. \tag{7.10}$$

Induction

$$\frac{\partial B_r^{(1)}}{\partial t} = B^{(0)}\frac{\partial V_r}{\partial z}, \tag{7.11}$$

$$\frac{\partial B_\varphi^{(1)}}{\partial t} = B^{(0)}\frac{\partial V_\varphi}{\partial z}, \tag{7.12}$$

$$\frac{\partial B_z^{(1)}}{\partial t} = -\frac{B^{(0)}}{r}\frac{\partial (rV_r)}{\partial r} = -B^{(0)}\frac{\partial V_r}{\partial r} - B^{(0)}\frac{V_r}{r}, \tag{7.13}$$

Pressure

$$\frac{\partial P^{(1)}}{\partial t} = -\gamma P^{(0)}\left(\frac{1}{r}\frac{\partial (rV_r)}{\partial r} + \frac{\partial V_z}{\partial z}\right) = -\gamma P^{(0)}\frac{\partial V_r}{\partial r} - \gamma P^{(0)}\frac{V_r}{r} - \gamma P^{(0)}\frac{\partial V_z}{\partial z}. \tag{7.14}$$

The system decouples into two independent subsystems. The φ-components (7.9) and (7.12) form a closed 2×2 system in $(V_\varphi, B_\varphi^{(1)})$ describing a pure shear Alfvén wave propagating along $\hat{e}_z$, consistent with subsection 5.1.1. The remaining five equations (7.14), (7.8), (7.10), (7.11), (7.13) form a coupled system in $(P^{(1)}, V_r, V_z, B_r^{(1)}, B_z^{(1)})$ describing the compressional (magnetosonic) branch.

7.2.2. Proof of Hyperbolicity

In general, hyperbolicity is charactereized by finite propagation speed.
To classify the reduced (r, z) system, we apply the definition of hyperbolicity for first-order systems from Godlewski and Raviart 2021 to the principal part of our balance law.

Since hyperbolicity is determined solely by the eigenvalues of the flux Jacobian $\mathsf{A}(\vec{u})$, and the geometric source term does not enter $\mathsf{A}(\vec{u})$, the classification proceeds identically to the source-free case Godlewski and Raviart 2021, Ch. VII, §1.1. Writing the system in the form

$$\frac{\partial \vec{u}}{\partial t} + \mathsf{A}_r \frac{\partial \vec{u}}{\partial r} + \mathsf{A}_z \frac{\partial \vec{u}}{\partial z} = \vec{s}(\vec{u}, \vec{r}), \tag{7.15}$$

where the state vector is

$$\vec{u} = \begin{pmatrix} V_r \\ V_z \\ B_r^{(1)} \\ B_z^{(1)} \\ P^{(1)} \end{pmatrix}, \tag{7.16}$$

the flux matrices and the source term are read off directly from (7.14), (7.8), (7.10), (7.11), (7.13):

$$\mathsf{A}_r = \begin{pmatrix} 0 & 0 & 0 & \frac{B^{(0)}}{\mu_0 \varrho^{(0)}} & \frac{1}{\varrho^{(0)}} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ B^{(0)} & 0 & 0 & 0 & 0 \\ \gamma P^{(0)} & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \mathsf{A}_z = \begin{pmatrix} 0 & 0 & -\frac{B^{(0)}}{\mu_0 \varrho^{(0)}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{\varrho^{(0)}} \\ -B^{(0)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \gamma P^{(0)} & 0 & 0 & 0 \end{pmatrix}, \tag{7.17}$$

$$\vec{s} = -\frac{V_r}{r} \begin{pmatrix} 0 \\ 0 \\ 0 \\ B^{(0)} \\ \gamma P^{(0)} \end{pmatrix}. \tag{7.18}$$

The system (7.15) is hyperbolic if and only if for all $\alpha_r, \alpha_z \in \mathbb{R}$ the combined matrix

$$\mathsf{A}(\alpha_r, \alpha_z) := \alpha_r \mathsf{A}_r + \alpha_z \mathsf{A}_z \tag{7.19}$$

has only real eigenvalues and is diagonalisable (Godlewski and Raviart 2021). If the eigenvalues are also unique, it is strictly hyperbolic. The eigenvalues λ are the characteristic speeds.
For $(\alpha_r, \alpha_z) = (0, 0)$ the problem is of course hyperbolic.
We get a characteristic polynomial of

$$\chi_A(\lambda) = \lambda \left[\lambda^4 - \left(\alpha_r^2 + \alpha_z^2\right) v_f^2 \lambda^2 + \alpha_z^2 \left(\alpha_r^2 + \alpha_z^2\right) c_s^2 v_A^2\right].$$

where

$$v_A = \frac{B^{(0)}}{\sqrt{\mu_0 \varrho^{(0)}}}, \qquad v_f = \sqrt{v_A^2 + c_s^2}, \qquad c_s = \sqrt{\frac{\gamma P^{(0)}}{\varrho^{(0)}}}. \tag{7.20}$$

The five eigenvalues are

$$
\begin{aligned}
\lambda_0 &= 0, \\
\pm\lambda_S &= \pm\sqrt{\frac{\alpha_r^2+\alpha_z^2}{2}\left(v_f^2 - \sqrt{v_f^4 - 4\frac{\alpha_z^2}{\alpha_r^2+\alpha_z^2}v_A^2c_s^2}\right)}, \\
\pm\lambda_F &= \pm\sqrt{\frac{\alpha_r^2+\alpha_z^2}{2}\left(v_f^2 + \sqrt{v_f^4 - 4\frac{\alpha_z^2}{\alpha_r^2+\alpha_z^2}v_A^2c_s^2}\right)}.
\end{aligned}
$$

which are real for all $\alpha_r, \alpha_z \in \mathbb{R}$ since $v_A, c_s, v_f \in \mathbb{R}$.
For $\alpha_r \neq 0, \alpha_z \neq 0$ all eigenvalues are different, their eigenvectors therefore independent and the system strictly hyperbolic. If $\alpha_r = 0$ and $v_A^2 = c_s^2$, which is realistic for plasma β of order unity, then $\pm\lambda_S = \pm\lambda_F$,and if $\alpha_z = 0$ also $\pm\lambda_S = \lambda_0 = 0$ In both cases the system remains hyperbolic.
The φ-system has eigenvalues $\pm\lambda_\varphi = \pm\alpha_z v_A$ and is thus hyperbolic if $\alpha_z = 0$ or $v_A = 0$, else it is strictly hyperbolic.
Summarizing we find that the (7.14)-(7.13) is a first order hyperbolic system of PDEs, and for most parameters even strictly hyperbolic.
We find that the fastest characteristic speed for any propagation is the maximum of λ_F at fixed $\alpha_r^2 + \alpha_z^2$, which is obtained at $\alpha_z = 0$ and has a value of $v_f\sqrt{\alpha_r^2 + \alpha_z^2}$. This corresponds to the fast magnetosonic wave discussed inplasma physics resources, here propagating perpendicular to the magnetic field. Thus v_f is the relevant speed governing the CFL criterion.
Since all characteristic speeds are real and finite, the system propagates information at bounded speeds, which is the defining property of a hyperbolic system and the necessary condition for an explicit time-marching scheme to be applicable.

7.2.3. FDTD Discretization

We will approximize the system of PDEs by finite difference equations, derived from first order Taylor expansion, as outlined by Yee in his paper *Numerical Solution of Initial Boundary Value Problems Involving Maxwell's Equations in Isotropic Media* (see Yee 1966), but for the (7.14)-(7.13) system of PDEs.

Staggered Grid

Due to our simplification of assuming no φ-dependence (subsection 7.2.1) our grid is only in the (r, z)-plane and because we assume $\vec{B}^{(0)} \parallel \hat{e}_z$, it only makes sense to consider the center regions of the Polomac without the arcs. Furthermore, we will restrict this discussion to the central cylinder. The outer vertical areas are analogue, but without the special attention to the $r = 0$ case.
The middle versions of (7.14) and (7.13) without the product rule yet applied are better suited for the concrete update equations due to the r factor coming in in both the nominator and the denominator. This also omits the source term.

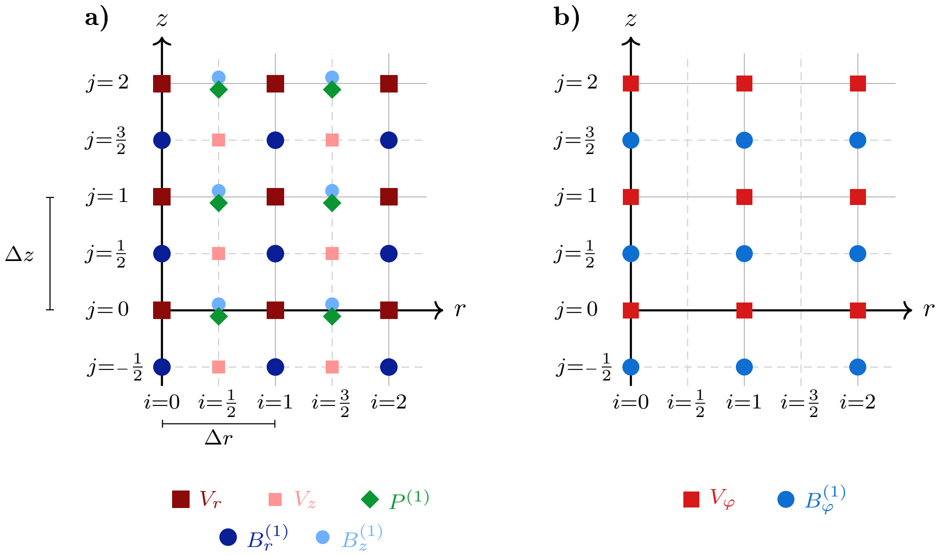

Figure 7.1.: Staggered grid layout for the linearised ideal MHD equations in the (r, z)-plane under axisymmetry ($\partial/\partial\varphi = 0$). Solid and dashed lines mark integer and half-integer grid positions, respectively. **(a)** Compressional branch: V_r and V_z are updated at half-integer timesteps $n+\frac{1}{2}$; $B_r^{(1)}$, $B_z^{(1)}$, and $P^{(1)}$ at integer timesteps $n+1$. Spatial nodes: V_r at $(i,\, j)$; V_z at $(i+\frac{1}{2},\, j+\frac{1}{2})$; $B_r^{(1)}$ at $(i,\, j+\frac{1}{2})$; $B_z^{(1)}$ and $P^{(1)}$ co-located at $(i+\frac{1}{2},\, j)$, offset vertically for visibility. Placing $P^{(1)}$ and $B_z^{(1)}$ at half-integer r-nodes avoids division by zero at $r = 0$. **(b)** Shear-Alfvén branch: V_φ (updated at $n+\frac{1}{2}$) at $(i,\, j)$; $B_\varphi^{(1)}$ (updated at $n+1$) at $(i,\, j+\frac{1}{2})$; decoupled from the compressional branch.

As in (7.14)-(7.13) we have the dependencies $P^{(1)}(\vec{V})$, $\vec{B}^{(1)}(\vec{V})$ and $\vec{V}(P^{(1)}, \vec{B}^{(1)})$, we choose to update $\vec{V}$ and $P^{(1)}, \vec{B}^{(1)}$ at staggered timesteps.
To avoid the zero division at $r = 0$, we place the variables $P^{(1)}, B_z^{(1)}$, whose update equations contain a factor $1/r$, at half-Δr nodes.

Discrete Update Equations

We then arrive at the discrete update equations for the compressional branch

$$
\begin{aligned}
\frac{[V_r]^{n+\frac{1}{2}}_{(i,j)} - [V_r]^{n-\frac{1}{2}}_{(i,j)}}{\Delta t} &= -\frac{1}{[\varrho^{(0)}]_{(i,j)}} \frac{[P^{(1)}]^n_{(i+\frac{1}{2},j)} - [P^{(1)}]^n_{(i-\frac{1}{2},j)}}{\Delta r} \\
&\quad + \frac{[B^{(0)}]_{(i,j)}}{\mu_0\,[\varrho^{(0)}]_{(i,j)}} \frac{[B_r^{(1)}]^n_{(i,j+\frac{1}{2})} - [B_r^{(1)}]^n_{(i,j-\frac{1}{2})}}{\Delta z} \\
&\quad - \frac{[B^{(0)}]_{(i,j)}}{\mu_0\,[\varrho^{(0)}]_{(i,j)}} \frac{[B_z^{(1)}]^n_{(i+\frac{1}{2},j)} - [B_z^{(1)}]^n_{(i-\frac{1}{2},j)}}{\Delta r}, \qquad (7.21) \\
\frac{[V_z]^{n+\frac{1}{2}}_{(i+\frac{1}{2},j+\frac{1}{2})} - [V_z]^{n-\frac{1}{2}}_{(i+\frac{1}{2},j+\frac{1}{2})}}{\Delta t} &= -\frac{1}{[\varrho^{(0)}]_{(i+\frac{1}{2},j+\frac{1}{2})}} \frac{[P^{(1)}]^n_{(i+\frac{1}{2},j+1)} - [P^{(1)}]^n_{(i+\frac{1}{2},j)}}{\Delta z}, \\
\frac{[B_r^{(1)}]^{n+1}_{(i,j+\frac{1}{2})} - [B_r^{(1)}]^n_{(i,j+\frac{1}{2})}}{\Delta t} &= [B^{(0)}]_{(i,j+\frac{1}{2})} \frac{[V_r]^{n+\frac{1}{2}}_{(i,j+1)} - [V_r]^{n+\frac{1}{2}}_{(i,j)}}{\Delta z}, \\
\frac{[B_z^{(1)}]^{(n+1)}_{(i+\frac{1}{2},j)} - [B_z^{(1)}]^{(n)}_{(i+\frac{1}{2},j)}}{\Delta t} &= -\frac{[B^{(0)}]_{(i+\frac{1}{2},j)}}{r_{i+\frac{1}{2}}} \frac{[rV_r]^{n+\frac{1}{2}}_{(i+1,j)} - [rV_r]^{n+\frac{1}{2}}_{(i,j)}}{\Delta r}, \\
\frac{[P^{(1)}]^{n+1}_{(i+\frac{1}{2},j)} - [P^{(1)}]^n_{(i+\frac{1}{2},j)}}{\Delta t} &= -\gamma [P^{(0)}]_{(i+\frac{1}{2},j)} \left(\frac{1}{r_{i+\frac{1}{2}}} \frac{[rV_r]^{n+\frac{1}{2}}_{(i+1,j)} - [rV_r]^{n+\frac{1}{2}}_{(i,j)}}{\Delta r} \right. \\
&\qquad \left. + \frac{[V_z]^{n+\frac{1}{2}}_{(i,j+\frac{1}{2})} - [V_z]^{n+\frac{1}{2}}_{(i,j-\frac{1}{2})}}{\Delta z} \right).
\end{aligned}
$$

For the shear Alfvén branch, we can choose the nodes freely and if we keep them analogous to the compression branch we find the discrete update equations

$$
\begin{aligned}
\frac{[V_\varphi]^{n+\frac{1}{2}}_{(i,j)} - [V_\varphi]^{n-\frac{1}{2}}_{(i,j)}}{\Delta t} &= \frac{[B^{(0)}]_{(i,j)}}{\mu_0\,[\varrho^{(0)}]_{(i,j)}} \frac{[B_\varphi^{(1)}]^n_{(i,j+\frac{1}{2})} - [B_\varphi^{(1)}]^n_{(i,j-\frac{1}{2})}}{\Delta z}, \\
\frac{[B_\varphi^{(1)}]^{n+1}_{(i,j+\frac{1}{2})} - [B_\varphi^{(1)}]^n_{(i,j+\frac{1}{2})}}{\Delta t} &= [B^{(0)}]_{(i,j+\frac{1}{2})} \frac{[V_\varphi]^{n+\frac{1}{2}}_{(i,j+1)} - [V_\varphi]^{n+\frac{1}{2}}_{(i,j)}}{\Delta z}.
\end{aligned}
$$

Furthermore we still have the constraint of no magnetic monopoles (7.1) which in discretized axisymmetric form reads

$$
\frac{1}{r_{i+\frac{1}{2}}} \frac{r_{i+1}\,[B_r^{(1)}]^n_{(i+1,\,j+\frac{1}{2})} - r_i\,[B_r^{(1)}]^n_{(i,\,j+\frac{1}{2})}}{\Delta r} + \frac{[B_z^{(1)}]^n_{(i+\frac{1}{2},\,j+1)} - [B_z^{(1)}]^n_{(i+\frac{1}{2},\,j)}}{\Delta z} = 0.
$$

Boundary Conditions

Let R be the radius of the central cylinder. We have the following boudary conditions.

Perfectly Conducting Wall As in subsection 4.2.2 we assume the walls to be perfectly conducting. Applying (4.17) we get $[B_r^{(1)}]_{i,j+\frac{1}{2}}^{(n)}|_{i\cdot\Delta r=R} = 0\ \forall n, j$.

No Plasma Flow through Wall This physical constraint implies $[V_r^{(1)}]_{i,j+\frac{1}{2}}^{(n)}|_{i\cdot\Delta r=R} = 0$ $\forall n, j$.

Axisymmetry Axisymmetry requires any quantity to obtain the same value at $0.5\Delta r$ as at $-0.5\Delta r$. Thus $\forall n$: $[P^{(1)}]^n_{(\frac{1}{2},j)} = [P^{(1)}]^n_{(-\frac{1}{2},j)}$ and $[B_z^{(1)}]^n_{(\frac{1}{2},j)} = [B_z^{(1)}]^n_{(-\frac{1}{2},j)}$ which implies that (7.21) at $r = 0$ takes the form

$$\frac{[V_r]^{n+\frac{1}{2}}_{(0,j)} - [V_r]^{n-\frac{1}{2}}_{(0,j)}}{\Delta t} = \frac{[B^{(0)}]_{(0,j)}}{\mu_0\ [\varrho^{(0)}]_{(0,j)}} \frac{[B_r^{(1)}]^n_{(0,j+\frac{1}{2})} - [B_r^{(1)}]^n_{(0,j-\frac{1}{2})}}{\Delta z}.$$

Courant-Friedrichs-Lewy Stability Condition

To obtain an initial estimate for a suitable timestep, the Courant-Friedrichs-Lewy (CFL) condition can be used. For explicit schemes, the CFL condition is a necessary, though not sufficient, criterion for stability, and requires that the numerical domain of dependence contains the physical domain of dependence (LeVeque 2002).
For our explicit scheme with two spatial dimensions and a maximum characteristic velocity of $v_f = \sqrt{v_A^2 + c_s^2}$ a safe bound is (Inan and Marshall 2011)

$$v_f \Delta t \leq \frac{1}{\sqrt{\frac{1}{(\Delta r)^2} + \frac{1}{(\Delta z)^2}}}. \tag{7.22}$$

We can use the Polomac target values from chapter 5 ($v_A \approx 5.5 \times 10^5\,\mathrm{m\,s^{-1}}$, $T_i \approx T_e = 100\,\mathrm{eV}$, $\gamma \sim 5/3$), keeping in mind that the adiabatic closure is an approximation and results explicitly dependent on γ ought to be treated with caution. Disregarding electrons' contribution to the mass density and treating the plasma as an ideal gas the adiabatic sound speed is (reformulation of $c_s = \sqrt{\gamma P_0/\varrho_0}$ from chapter 8 of Freidberg 2014)

$$c_s = \sqrt{\frac{\gamma k_B T}{m_i}} \approx 1.3 \times 10^5\,\mathrm{m\,s^{-1}},$$

giving $v_f = \sqrt{v_A^2 + c_s^2} \approx 5.6 \times 10^5\,\mathrm{m\,s^{-1}}$.
For a square grid $\Delta r = \Delta z = \Delta x$, (7.22) simplifies to $\Delta t \leq \Delta x/(\sqrt{2} v_f)$. For $L = R = 0.15\,\mathrm{m}$ resolved by 100 gridpoints this implies $\Delta t \lesssim 1.9 \times 10^{-9}\,\mathrm{s}$. This is two orders of magnitude smaller than the Alfvén timescale $\tau_A \approx 5.5 \times 10^{-7}\,\mathrm{s}$, so a simulation spanning several Alfvén times would require of order $10^3 - 10^4$ timesteps.
A more rigorous stability assessment would require a von Neumann analysis of the amplification factors of the discretized equations or an analysis of boundedness in a suitable

discrete energy norm. Such analyses are not performed here, the CFL criterion is instead used as a practical timestep restriction.

7.2.4. Verification Tests

A fully implemented solver should pass three classes of consistency checks, which are stated here for completeness.

Wave Propagation For a uniform equilibrium, the linearized system admits exact plane-wave solutions with dispersion relations given by (7.20) (Freidberg 2014). A numerical solution initialized as a single Fourier mode should reproduce the correct phase velocity ω/k to second order in Δr, Δz, and Δt, and the wave amplitude should remain constant over many Alfvén times.

Energy Conservation Ideal MHD conserves the total energy Freidberg 2014, Ch. 8.

$\nabla \cdot \vec{B}$ Control The constraint $\nabla \cdot \vec{B}^{(1)} = 0$ (7.1) is preserved analytically by (7.4) and, by construction, also by its discrete counterpart.

8. Conclusion, Outlook and Acknowledgements

8.1. Conclusion

This work developed a self-contained theoretical MHD framework for the Polomac concept, proceeding from first principles to a concrete numerical implementation plan.

Ideal MHD Equations (Chapter 2) Starting from the Boltzmann equation, velocity moments were taken to obtain species-fluid equations, which were then combined into a single-fluid model. The ideal MHD limit was obtained by imposing six assumptions (quasi-neutrality, negligible electron inertia, negligible displacement current, zero resistivity, small Larmor radius, and adiabatic isotropic closure), each verified quantitatively for Polomac parameters. The resulting system consists of a mass-continuity equation, a momentum equation, the ideal Ohm's law, the adiabatic closure, and the two relevant Maxwell equations.

Polomac Geometry (Chapter 3). The magnetic field is predominantly poloidal, the toroidal component satisfies $B_\varphi/|\vec{B}_{\mathrm{pol}}| \equiv \varepsilon \ll 1$ and enters the MHD equations only as an $\mathcal{O}(\varepsilon)$ correction. To leading order $B_\varphi \equiv 0$, a simplification used throughout all subsequent chapters. Axisymmetry is exact in the arc sections and broken by the four magnetic tunnels in the midplane.

Static Equilibrium (Chapter 4). Setting $\vec{V} = 0$, $B_\varphi = 0$, and $\partial/\partial\varphi = 0$ reduces the force-balance problem to the poloidal flux function $\psi(r, z)$ governed by the reduced Grad–Shafranov equation (4.16). Boundary conditions are regularity ($\nabla\psi \to 0$ at $r = 0$) and $\psi = \mathrm{const}$ on the conducting walls. The pressure profile $P(\psi)$ is not determined by ideal MHD and must be supplied by transport theory or experimental data.

Characteristic Parameters (Chapter 5). Three dimensionless parameters govern the MHD dynamics of the Polomac:

- **Alfvén velocity.** $v_A \approx 5.5 \times 10^5\,\mathrm{m\,s^{-1}}$, giving $v_A/c \approx 1.8 \times 10^{-3} \ll 1$, which justifies neglecting the displacement current.
- **Lundquist number.** $S \in [2.8 \times 10^4, 2.1 \times 10^5] \gg 1$, confirming that resistive diffusion is negligible on MHD timescales and the ideal limit $\eta \to 0$ is appropriate.

- **Plasma β.** $\beta \in [0.04,\ 1.01]$ if the upper range is achievable, the Polomac could confine fusion-grade pressures with a modest magnetic field. These are design targets and have not been experimentally verified.

Geometry Comparison (Chapter 6). The Polomac drives no toroidal plasma current, so current-driven kink and tearing instabilities—and the associated risk of major disruptions—have no analogue here, unlike in the tokamak. The principal MHD stability threat is the interchange instability: with identically zero rotational transform there is no magnetic shear, and the geometry does not provide average good curvature; the only stabilizing mechanism is plasma compressibility, as demonstrated experimentally in LDX (Garnier, Hansen, et al. 2006). A Polomac-specific risk is the hot-electron interchange instability: depending on how well the four magnetic tunnels breaking toroidal axisymmetrywork, they may act analogously to the rod supports that triggered HEI bursts in the non-levitated LDX operating mode. Whether compressibility stabilization survives this four-fold symmetry breaking has not yet been quantitatively assessed (Elio et al. 2024).

FDTD Framework (Chapter 7). The ideal MHD equations were linearized around a uniform static equilibrium $\vec{B}^{(0)} = B^{(0)}\hat{e}_z$, reduced to 2D axisymmetric cylindrical coordinates, and shown to form a first-order hyperbolic system with characteristic speeds $0, v_A, c_s, v_f\}$, where $v_f = \sqrt{v_A^2 + c_s^2}$ is the fast magnetosonic speed (Freidberg 2014). A staggered-grid (Yee-type) discretization was derived, separating a decoupled shear-Alfvén branch $(V_\varphi, B_\varphi^{(1)})$ from the compressional branch $(V_r, V_z, B_r^{(1)}, B_z^{(1)}, P^{(1)})$. The CFL stability bound is $\Delta t \leq \Delta x/(\sqrt{2}\, v_f)$, requiring $\sim \mathcal{O}(10^3 - 10^4)$ timesteps per Alfvén time for a 100-point grid at Polomac parameters.

8.2. Outlook

8.2.1. Numerical Implementation

The immediate next step is to implement the FDTD solver specified in Chapter 7. Correctness should be established through the three verification tests described in Section 7.2.4: (i) a plane-wave test confirming that the shear-Alfvén and fast-magnetosonic dispersion relations from subsection 7.2.2 are reproduced to second order globally in Δr, Δz, Δt; (ii) conservation of the discrete total energy over many Alfvén times; and (iii) machine-precision preservation of the discrete $\nabla \cdot \vec{B}^{(1)} = 0$ constraint (7.1).

8.2.2. Equilibrium beyond the Uniform Background

The FDTD framework currently linearizes around a uniform $B^{(0)}\hat{e}_z$ equilibrium. A natural and physically more relevant next step is to use a Grad–Shafranov solution fulfilling (4.16) with an experimentally verified or from transport theory derived pressure profile $P(\psi)$ as the equilibrium state and linearize around it.

8.2.3. Extension to Non-Axisymmetric Perturbations

The solver framework assumes axisymmetry ($\partial/\partial\varphi = 0$), corresponding to the $m = 0$ toroidal mode. Since the equilibrium is axisymmetric, perturbations of arbitrary toroidal mode number m can be incorporated without increasing the dimensionality of the numerical problem. Writing every perturbation field as

$$Q_1(r, \varphi, z, t) = \hat{Q}_1(r, z, t)\, e^{im\varphi}, \qquad m \in \mathbb{Z}, \tag{8.1}$$

replaces $\partial/\partial\varphi \to im$ everywhere in the linearized equations (7.2)-(7.5). Equation (8.1) is inspired by the Ansatz for the θ-pinch from section 11.3 of Freidberg 2014. Each mode number m then yields an independent 2D problem on the same (r, z) grid, with additional im/r terms appearing in the divergence and curl operators. The solver structure, CFL condition, and boundary conditions should carry over with only minor modifications; the one qualitative change is that V_φ and $B_\varphi^{(1)}$ no longer decouple from the poloidal components for $m \neq 0$, so all six field components must be evolved simultaneously. At the axis $r = 0$ the regularity condition becomes mode-dependent: fields must vanish there for $m \geq 1$, because they need to be the same for all φ.

Surveying the lowest few mode numbers is of direct physical interest given the closed-field-line geometry discussed in Section 8.1. The four magnetic tunnels break axisymmetry with four-fold symmetry, possibly coupling $m = 4k$ and their harmonics and destabilizing the fields.

8.2.4. Resources and Collaboration

A natural starting point for further numerical work is exploring the resources already established by LDX and RT-1.

The OpenStar project is currently developing a levitated-dipole reactor concept in a regime closely related to the Polomac. Possibly an exchange of numerical methods and stability analysis, as well as other collaboration, would be mutually beneficial.

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Bibliography

Beresnyak, A. (2023). *NRL Plasma Formulary*. Tech. rep. Radiation Hydrodynamics Branch, Plasma Physics Division. Supported by the Office of Naval Research. Washington, DC 20375: Naval Research Laboratory. URL: `https://www.nrl.navy.mil/Portals/38/PDF%20Files/NRL_Plasma_Formulary_2023.pdf`.

Blandford, Roger D. and Kip S. Thorne (2013). "Applications of Classical Physics". In: chap. 20. URL: `https://api.semanticscholar.org/CorpusID:118651591`.

Chen, Francis F. (2016). *Introduction to Plasma Physics and Controlled Fusion*. 3rd ed. Springer Cham, pp. xii+490. ISBN: 978-3-319-22309-4. DOI: `10.1007/978-3-319-22309-4`.

Elio, F. et al. (2024). "Technical Report: The Polomac approach to fusion energy". In: *The Journal of Technological and Space Plasmas* 5.1, pp. 172–180. DOI: `10.31281/med9bh43`. URL: `https://jtsp.eu/jtsp/article/view/32`.

Fasoli, A. et al. (2018). *Plasma Physics: Introduction*. `https://www.openlearnity.org/courses/plasmaintroduction/`. Online course by École polytechnique fédérale de Lausanne, accessed Spring 2026.

Freidberg, Jeffrey P. (2014). *Ideal MHD*. Cambridge University Press.

Garnier, D. T., A. Hansen, et al. (May 2006). "Production and study of high-beta plasma confined by a superconducting dipole magnet". In: *Physics of Plasmas* 13.5, p. 056111. ISSN: 1070-664X. DOI: `10.1063/1.2186616`. eprint: `https://pubs.aip.org/aip/pop/article-pdf/doi/10.1063/1.2186616/15804515/056111_1_online.pdf`. URL: `https://doi.org/10.1063/1.2186616`.

Garnier, D. T., J. Kesner, and M. E. Mauel (Sept. 1999). "Magnetohydrodynamic stability in a levitated dipole". In: *Physics of Plasmas* 6.9, pp. 3431–3434. ISSN: 1070-664X. DOI: `10.1063/1.873601`. eprint: `https://pubs.aip.org/aip/pop/article-pdf/6/9/3431/19026255/3431_1_online.pdf`. URL: `https://doi.org/10.1063/1.873601`.

Godlewski, Edwige and Pierre-Arnaud Raviart (2021). *Numerical Approximation of Hyperbolic Systems of Conservation Laws*. 2nd ed. Vol. 118. Applied Mathematical Sciences. New York, NY: Springer. ISBN: 978-1-0716-1344-3. DOI: `10.1007/978-1-0716-1344-3`.

Goldston, Robert J. and Paul H. Rutherford (1995). *Introduction to Plasma Physics*. Bristol and Philadelphia: Institute of Physics Publishing, p. 128. ISBN: 0-7503-0325-5.

Grulke, Olaf et al. (2026). "Overview of Wendelstein 7-X high-performance operation". In: *Nuclear Fusion*. Accepted manuscript published online 14 April 2026. DOI: `10.1088/1741-4326/ae5f32`.

Inan, Umran S. and Robert A. Marshall (2011). "Numerical stability of finite difference methods". In: *Numerical Electromagnetics: The FDTD Method*. Cambridge University Press, pp. 113–131.

LeVeque, Randall J. (2002). *Finite Volume Methods for Hyperbolic Problems.* Cambridge Texts in Applied Mathematics. Cambridge University Press.

Nishiura, M. et al. (2015). "Improved beta (local beta > 1) and density in electron cyclotron resonance heating on the RT-1 magnetosphere plasma". In: *Nuclear Fusion* 55.5, p. 053019. DOI: 10.1088/0029-5515/55/5/053019.

Rosenbluth, M.N and C.L Longmire (1957). "Stability of plasmas confined by magnetic fields". In: *Annals of Physics* 1.2, pp. 120–140. ISSN: 0003-4916. DOI: https://doi.org/10.1016/0003-4916(57)90055-6. URL: https://www.sciencedirect.com/science/article/pii/0003491657900556.

Simpson, T. et al. (2026). *Deuterium-Tritium Levitated Dipole Fusion Power Plants.* arXiv: 2602.20564 [physics.plasm-ph]. URL: https://arxiv.org/abs/2602.20564.

Sips, A. C. C. et al. (2018). "Assessment of the baseline scenario at $q_{95} \sim 3$ for ITER". In: *Nuclear Fusion* 58.12. DOI: 10.1088/1741-4326/aade57.

Yee, Kane (1966). "Numerical solution of initial boundary value problems involving maxwell's equations in isotropic media". In: *IEEE Transactions on Antennas and Propagation* 14.3, pp. 302–307. DOI: 10.1109/TAP.1966.1138693.

A. Appendices

A.1. Species Momentum Equation Derivation

Choose $g_s(\vec{v}) = m_s\vec{v}$ and take the momentum moment of the Boltzmann equation:

$$\Rightarrow \int d\vec{v}\, m_s\vec{v} \left[\frac{\partial f_s}{\partial t} + \vec{v}\cdot\frac{\partial f_s}{\partial \vec{r}} + \frac{q_s}{m_s}(\vec{E}^{lr} + \vec{v}\times\vec{B}^{lr})\cdot\frac{\partial f_s}{\partial \vec{v}} - \left(\frac{\partial f}{\partial t}\right)_{coll}\right] = 0.$$

This gives us four terms to evaluate:

$$\begin{aligned}
&\text{Term 1:}\ \int d\vec{v}\, m_s\vec{v}\frac{\partial f_s}{\partial t}\\
&\text{Term 2:}\ \int d\vec{v}\, m_s\vec{v}(\vec{v}\cdot\nabla_{\vec{r}} f_s)\\
&\text{Term 3:}\ \int d\vec{v}\, m_s\vec{v}\left[\frac{q_s}{m_s}(\vec{E} + \vec{v}\times\vec{B})\cdot\nabla_{\vec{v}} f_s\right]\\
&\text{Term 4:}\ -\int d\vec{v}\, m_s\vec{v}\left(\frac{\partial f}{\partial t}\right)_{coll}
\end{aligned}$$

Term 1: Time derivative

$$\begin{aligned}
\text{Term 1} &= \int d^3v\, m_s\vec{v}\frac{\partial f_s}{\partial t}\\
&= m_s\frac{\partial}{\partial t}\int d^3v\, \vec{v} f_s \quad (\vec{v} \text{ independent of } t)\\
&= m_s\frac{\partial}{\partial t}(n_s\vec{u}_s)\\
&= m_s\left[\frac{\partial n_s}{\partial t}\vec{u}_s + n_s\frac{\partial \vec{u}_s}{\partial t}\right]
\end{aligned}$$

Term 2: Spatial advection

$$\begin{aligned}
\text{Term 2} &= \int d^3v\, m_s\vec{v}(\vec{v}\cdot\nabla_{\vec{r}} f_s)\\
&= m_s\int d^3v\, v_i v_j\frac{\partial f_s}{\partial r_j} \quad \text{(component form)}\\
&= m_s\frac{\partial}{\partial r_j}\int d^3v\, v_i v_j f_s \quad (\vec{v} \text{ independent of } \vec{r})
\end{aligned}$$

We now evaluate the integral $\int d^3v\, v_i v_j f_s$ using the velocity decomposition:

$$\vec{v} = \vec{u}_s + \vec{w}, \quad \text{where } \vec{w} = \vec{v} - \vec{u}_s.$$

Then:

$$\begin{aligned} v_i v_j &= (u_{s,i} + w_i)(u_{s,j} + w_j) \\ &= u_{s,i}u_{s,j} + u_{s,i}w_j + w_i u_{s,j} + w_i w_j. \end{aligned}$$

Integrating term by term:

$$\begin{aligned} \int d^3v\, u_{s,i}u_{s,j} f_s &= u_{s,i}u_{s,j} n_s, \\ \int d^3v\, u_{s,i} w_j f_s &= u_{s,i} \int d^3v\, (v_j - u_{s,j}) f_s = u_{s,i}[n_s u_{s,j} - n_s u_{s,j}] = 0, \\ \int d^3v\, w_i u_{s,j} f_s &= 0 \quad \text{(similarly)}, \\ \int d^3v\, w_i w_j f_s &= \frac{P_{s,ij}}{m_s}, \end{aligned}$$

where the pressure tensor is defined as:

$$P_{s,ij} \equiv m_s \int d^3v\, (v_i - u_{s,i})(v_j - u_{s,j}) f_s.$$

Therefore:

$$\int d^3v\, v_i v_j f_s = n_s u_{s,i} u_{s,j} + \frac{P_{s,ij}}{m_s}.$$

Substituting back into Term 2:

$$\begin{aligned} \text{Term 2} &= m_s \frac{\partial}{\partial r_j} \left[n_s u_{s,i} u_{s,j} + \frac{P_{s,ij}}{m_s} \right] \\ &= m_s \frac{\partial}{\partial r_j} (n_s u_{s,i} u_{s,j}) + \frac{\partial P_{s,ij}}{\partial r_j}. \end{aligned}$$

Expanding the first term using the product rule:

$$\begin{aligned} \frac{\partial}{\partial r_j}(n_s u_{s,i} u_{s,j}) &= n_s u_{s,j} \frac{\partial u_{s,i}}{\partial r_j} + u_{s,i} \frac{\partial}{\partial r_j}(n_s u_{s,j}) \\ &= n_s (\vec{u}_s \cdot \nabla_{\vec{r}}) u_{s,i} + u_{s,i} \nabla_{\vec{r}} \cdot (n_s \vec{u}_s). \end{aligned}$$

In vector notation:

$$\text{Term 2} = m_s n_s (\vec{u}_s \cdot \nabla_{\vec{r}}) \vec{u}_s + m_s \vec{u}_s [\nabla_{\vec{r}} \cdot (n_s \vec{u}_s)] + \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s.$$

Combining Terms 1 and 2

Adding Terms 1 and 2:

$$\begin{aligned}\text{Term 1} + \text{Term 2} &= m_s \frac{\partial n_s}{\partial t}\vec{u}_s + m_s n_s \frac{\partial \vec{u}_s}{\partial t} \\ &\quad + m_s n_s (\vec{u}_s \cdot \nabla_{\vec{r}})\vec{u}_s + m_s \vec{u}_s [\nabla_{\vec{r}} \cdot (n_s \vec{u}_s)] + \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s\end{aligned}$$

Regrouping:

$$= m_s n_s \left[\frac{\partial \vec{u}_s}{\partial t} + (\vec{u}_s \cdot \nabla_{\vec{r}})\vec{u}_s\right] + m_s \vec{u}_s \left[\frac{\partial n_s}{\partial t} + \nabla_{\vec{r}} \cdot (n_s \vec{u}_s)\right] + \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s.$$

Using the species continuity equation:

$$\frac{\partial n_s}{\partial t} + \nabla_{\vec{r}} \cdot (n_s \vec{u}_s) = 0,$$

the second term vanishes, leaving:

$$\text{Term 1} + \text{Term 2} = m_s n_s \left[\frac{\partial \vec{u}_s}{\partial t} + (\vec{u}_s \cdot \nabla_{\vec{r}})\vec{u}_s\right] + \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s.$$

Using the *convective derivative*:

$$\frac{d\vec{u}_s}{dt} \equiv \frac{\partial \vec{u}_s}{\partial t} + (\vec{u}_s \cdot \nabla_{\vec{r}})\vec{u}_s,$$

we obtain:

$$\text{Term 1} + \text{Term 2} = m_s n_s \frac{d\vec{u}_s}{dt} + \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s.$$

Term 3: Lorentz force

$$\begin{aligned}\text{Term 3} &= \int d^3v\, m_s \vec{v} \left[\frac{q_s}{m_s}(\vec{E} + \vec{v} \times \vec{B}) \cdot \nabla_{\vec{v}} f_s\right] \\ &= q_s \int d^3v\, \vec{v} \cdot [(\vec{E} + \vec{v} \times \vec{B}) \cdot \nabla_{\vec{v}} f_s].\end{aligned}$$

In component form:

$$[\text{Term 3}]_i = q_s \int d^3v\, v_i [(E_j + (v \times B)_j) \frac{\partial f_s}{\partial v_j}].$$

We use integration by parts. From the product rule:

$$\frac{\partial}{\partial v_j}[v_i (E_j + (v \times B)_j) f_s] = v_i (E_j + (v \times B)_j) \frac{\partial f_s}{\partial v_j} + f_s \frac{\partial}{\partial v_j}[v_i (E_j + (v \times B)_j)].$$

Rearranging:

$$v_i(E_j + (v \times B)_j)\frac{\partial f_s}{\partial v_j} = \frac{\partial}{\partial v_j}[v_i(E_j + (v \times B)_j)f_s] - f_s\frac{\partial}{\partial v_j}[v_i(E_j + (v \times B)_j)].$$

Integrating over all velocity space:

$$\int d^3v\, v_i(E_j+(v\times B)_j)\frac{\partial f_s}{\partial v_j} = \underbrace{\int d^3v\, \frac{\partial}{\partial v_j}[v_i(E_j + (v \times B)_j)f_s]}_{=0} - \int d^3v\, f_s\frac{\partial}{\partial v_j}[v_i(E_j+(v\times B)_j)].$$

The first integral vanishes by the divergence theorem (surface integral at $|\vec{v}| \to \infty$ where $f_s \to 0$). For the second integral, we compute:

$$\begin{aligned}\frac{\partial}{\partial v_j}[v_i(E_j + (v \times B)_j)] &= \frac{\partial v_i}{\partial v_j}(E_j + (v \times B)_j) + v_i\frac{\partial}{\partial v_j}(E_j + (v \times B)_j) \\ &= \delta_{ij}(E_j + (v \times B)_j) + v_i\frac{\partial}{\partial v_j}(E_j + (v \times B)_j).\end{aligned}$$

The first term gives:

$$\delta_{ij}(E_j + (v \times B)_j) = E_i + (v \times B)_i.$$

For the second term, note that

- $\frac{\partial E_j}{\partial v_j} = 0 \quad (\vec{E}$ independent of $\vec{v})$
- $\frac{\partial}{\partial v_j}(v \times B)_j = \frac{\partial}{\partial v_j}(\varepsilon_{jkl}v_kB_l) = \varepsilon_{jkl}B_l\delta_{jk} = \varepsilon_{jjl}B_l = 0$

The last equality holds because $(v \times B)_j$ is independent of v_j (the cross product structure ensures the j-th component depends only on the other velocity components).
Therefore:

$$\frac{\partial}{\partial v_j}[v_i(E_j + (v \times B)_j)] = E_i + (v \times B)_i.$$

Substituting back:

$$\begin{aligned}[\text{Term 3}]_i &= -q_s\int d^3v\, f_s[E_i + (v \times B)_i] \\ &= -q_sE_i\int d^3v\, f_s - q_s\int d^3v\, f_s(v \times B)_i \\ &= -q_sn_sE_i - q_s\int d^3v\, f_s\varepsilon_{ijk}v_jB_k \\ &= -q_sn_sE_i - q_s\varepsilon_{ijk}B_k\int d^3v\, v_jf_s \\ &= -q_sn_sE_i - q_s\varepsilon_{ijk}B_k(n_su_{s,j}) \\ &= -q_sn_s[E_i + (u_s \times B)_i].\end{aligned}$$

In vector notation:

$$\text{Term 3} = -q_sn_s(\vec{E} + \vec{u}_s \times \vec{B}).$$

Term 4: Collision term

$$\text{Term 4} = -\int d^3v\, m_s\vec{v}\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \equiv \vec{R}_s.$$

We can rewrite this using the velocity decomposition $\vec{v} = \vec{u}_s + \vec{w}$:

$$\begin{aligned}\vec{R}_s &= -\int d^3v\, m_s(\vec{u}_s + \vec{w})\left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}} \\ &= -m_s\vec{u}_s\int d^3v\,\left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}} - m_s\int d^3v\,\vec{w}\left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}}.\end{aligned}$$

The first integral vanishes due to particle conservation in collisions:

$$\int d^3v\,\left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}} = 0.$$

Therefore, both forms are equivalent:

$$\vec{R}_s = -\int d^3v\, m_s\vec{v}\left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}} = -\int d^3v\, m_s(\vec{v} - \vec{u}_s)\left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}}.$$

Physical interpretation: $\vec{R}_s$ represents the momentum transferred to species s from other species through collisions.
Properties:

- For collisions within the same species: $\vec{R}_s = 0$ (momentum conserved)
- For multi-species plasma: $\sum_s \vec{R}_s = 0$ (total momentum conserved)

Final Result: Species Momentum equation

Combining all terms (Term 1 + Term 2 + Term 3 + Term 4 = 0):

$$m_s n_s \frac{d\vec{u}_s}{dt} + \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s - q_s n_s(\vec{E} + \vec{u}_s \times \vec{B}) + \vec{R}_s = 0.$$

Rearranging:

$$m_s n_s \frac{d\vec{u}_s}{dt} = q_s n_s(\vec{E} + \vec{u}_s \times \vec{B}) - \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s + \vec{R}_s,$$

where $\frac{d}{dt} = \frac{\partial}{\partial t} + \vec{u}_s \cdot \nabla_{\vec{r}}$ is the convective derivative.

A.2. Derivation of the Single Fluid Momentum Equation

This section presents a complete derivation of the single fluid momentum equation starting from the species momentum equations obtained from kinetic theory.

Starting Point: Species Momentum Equations

For each species s (electrons and ions), the momentum equation (2.6) derived from the Boltzmann kinetic equation is

$$m_s n_s \frac{d\vec{u}_s}{dt} = q_s n_s(\vec{E} + \vec{u}_s \times \vec{B}) - \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s + \vec{R}_s,$$

where:

- $\dfrac{d}{dt} = \dfrac{\partial}{\partial t} + \vec{u}_s \cdot \nabla_{\vec{r}}$ is the convective derivative for species s
- $\overleftrightarrow{P}_s = m_s \displaystyle\int d^3v\, (\vec{v} - \vec{u}_s) \otimes (\vec{v} - \vec{u}_s) f_s$ is the species-frame pressure tensor
- $\vec{R}_s = \displaystyle\int d^3v\, m_s(\vec{v} - \vec{u}_s) \left(\frac{\partial f_s}{\partial t}\right)_{\text{coll}}$ is the collisional momentum transfer

Summation of Species Equations

Summing the momentum equations over all species:

$$\sum_s m_s n_s \frac{d\vec{u}_s}{dt} = \sum_s q_s n_s(\vec{E} + \vec{u}_s \times \vec{B}) - \sum_s \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s + \sum_s \vec{R}_s.$$

The right-hand side simplifies as follows:

Lorentz force terms:

$$\sum_s q_s n_s \vec{E} = \vec{E} \sum_s q_s n_s = \rho \vec{E}$$

$$\sum_s q_s n_s (\vec{u}_s \times \vec{B}) = \left(\sum_s q_s n_s \vec{u}_s\right) \times \vec{B} = \vec{J} \times \vec{B}$$

Collision term:

$$\sum_s \vec{R}_s = 0,$$

since total momentum is conserved in collisions.
Thus,

$$\sum_s m_s n_s \frac{d\vec{u}_s}{dt} = \rho \vec{E} + \vec{J} \times \vec{B} - \sum_s \nabla_{\vec{r}} \cdot \overleftrightarrow{P}_s.$$

Single Fluid Variables

We define the species drift velocity relative to the total CM velocity $\vec{V}$

$$\vec{w}_s = \vec{u}_s - \vec{V} \quad \text{(species drift from CM)},$$

A crucial property of the drift velocity is:

$$\sum_s m_s n_s \vec{w}_s = \sum_s m_s n_s (\vec{u}_s - \vec{V}) = \varrho\vec{V} - \varrho\vec{V} = 0. \tag{A.1}$$

The Key Identity

We now prove that the sum of species convective derivatives can be rewritten as:

$$\sum_s m_s n_s \frac{d\vec{u}_s}{dt} = \varrho\frac{d\vec{V}}{dt} + \nabla_{\vec{r}} \cdot \left[\sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s\right]. \tag{A.2}$$

Working in component form, we have:

$$\sum_s m_s n_s \frac{du_{s,i}}{dt} = \sum_s m_s n_s \left[\frac{\partial u_{s,i}}{\partial t} + u_{s,j}\frac{\partial u_{s,i}}{\partial r_j}\right].$$

Substituting $u_{s,i} = V_i + w_{s,i}$:

$$= \sum_s m_s n_s \left[\frac{\partial V_i}{\partial t} + \frac{\partial w_{s,i}}{\partial t} + V_j\frac{\partial V_i}{\partial r_j} + V_j\frac{\partial w_{s,i}}{\partial r_j} + w_{s,j}\frac{\partial V_i}{\partial r_j} + w_{s,j}\frac{\partial w_{s,i}}{\partial r_j}\right].$$

The terms involving only V give:

$$\sum_s m_s n_s \left[\frac{\partial V_i}{\partial t} + V_j\frac{\partial V_i}{\partial r_j}\right] = \varrho\frac{dV_i}{dt}.$$

For the remaining terms, we use the product rule:

$$m_s n_s \frac{\partial w_{s,i}}{\partial t} = \frac{\partial}{\partial t}(m_s n_s w_{s,i}) - w_{s,i}\frac{\partial(m_s n_s)}{\partial t}.$$

Since $\sum_s m_s n_s w_{s,i} = 0$ for all t and $\vec{r}$ (equation A.1), its time and spatial derivatives also vanish:

$$\sum_s \frac{\partial}{\partial t}(m_s n_s w_{s,i}) = 0, \quad \sum_s \frac{\partial}{\partial r_j}(m_s n_s w_{s,i}) = 0.$$

From the species continuity equation $\frac{\partial(m_s n_s)}{\partial t} + \nabla \cdot (m_s n_s \vec{u}_s) = 0$, we obtain:

$$\frac{\partial(m_s n_s)}{\partial t} + V_j\frac{\partial(m_s n_s)}{\partial r_j} = -m_s n_s\frac{\partial u_{s,j}}{\partial r_j} - w_{s,j}\frac{\partial(m_s n_s)}{\partial r_j}.$$

Using $\frac{\partial u_{s,j}}{\partial r_j} = \frac{\partial V_j}{\partial r_j} + \frac{\partial w_{s,j}}{\partial r_j}$ and the fact that $\sum_s m_s n_s w_{s,i} = 0$, we find that the remaining terms combine to give:

$$\sum_s \left[m_s n_s w_{s,i} \frac{\partial w_{s,j}}{\partial r_j} + m_s n_s w_{s,j} \frac{\partial w_{s,i}}{\partial r_j} + w_{s,i} w_{s,j} \frac{\partial (m_s n_s)}{\partial r_j} \right].$$

Applying the product rule in reverse:

$$\frac{\partial}{\partial r_j}(m_s n_s w_{s,i} w_{s,j}) = w_{s,i} w_{s,j} \frac{\partial (m_s n_s)}{\partial r_j} + m_s n_s w_{s,i} \frac{\partial w_{s,j}}{\partial r_j} + m_s n_s w_{s,j} \frac{\partial w_{s,i}}{\partial r_j},$$

we obtain:

$$= \sum_s \frac{\partial}{\partial r_j}(m_s n_s w_{s,i} w_{s,j}) = \frac{\partial}{\partial r_j} \left[\sum_s m_s n_s w_{s,i} w_{s,j} \right].$$

This completes the proof of equation (A.2).

Pressure Tensor Relationships

We define the center-of-mass frame pressure contribution for species s as:

$$\overleftrightarrow{P}_s^{CM} = m_s \int d^3v \, (\vec{v} - \vec{V}) \otimes (\vec{v} - \vec{V}) f_s.$$

To relate this to the species-frame pressure, we decompose:

$$\vec{v} - \vec{V} = (\vec{v} - \vec{u}_s) + (\vec{u}_s - \vec{V}) = \vec{w} + \vec{w}_s,$$

where $\vec{w} = \vec{v} - \vec{u}_s$ is the thermal velocity. Expanding:

$$\overleftrightarrow{P}_s^{CM} = m_s \int d^3v \, (\vec{w} + \vec{w}_s) \otimes (\vec{w} + \vec{w}_s) f_s.$$

This yields four terms:

- **Term 1:** $m_s \int d^3v \, \vec{w} \otimes \vec{w} \, f_s = \overleftrightarrow{P}_s$
- **Term 2:** $m_s \int d^3v \, \vec{w} \otimes \vec{w}_s \, f_s = \vec{w}_s \otimes m_s \int d^3v \, \vec{w} f_s = 0$
- **Term 3:** $m_s \int d^3v \, \vec{w}_s \otimes \vec{w} \, f_s = 0$
- **Term 4:** $m_s \int d^3v \, \vec{w}_s \otimes \vec{w}_s \, f_s = m_s n_s \vec{w}_s \otimes \vec{w}_s$

The mixed terms (2 and 3) vanish because $\int d^3v \, \vec{w} f_s = 0$ by definition of the mean velocity.
Thus,

$$\overleftrightarrow{P}_s^{CM} = \overleftrightarrow{P}_s + m_s n_s \vec{w}_s \otimes \vec{w}_s.$$

Summing over all species:

$$\overleftrightarrow{P}^{CM} \equiv \sum_s \overleftrightarrow{P}_s^{CM} = \sum_s \overleftrightarrow{P}_s + \sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s. \tag{A.3}$$

Rearranging:

$$\sum_s \overleftrightarrow{P}_s = \overleftrightarrow{P}^{CM} - \sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s.$$

Final Assembly

Combining equation (A.2) with equation (A.2):

$$\varrho \frac{d\vec{V}}{dt} + \nabla_{\vec{r}} \cdot \left[\sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s\right] = \rho \vec{E} + \vec{J} \times \vec{B} - \nabla_{\vec{r}} \cdot \left[\sum_s \overleftrightarrow{P}_s\right].$$

Using equation (A.3):

$$\nabla_{\vec{r}} \cdot \left[\sum_s \overleftrightarrow{P}_s\right] = \nabla_{\vec{r}} \cdot \overleftrightarrow{P}^{CM} - \nabla_{\vec{r}} \cdot \left[\sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s\right].$$

Substituting:

$$\begin{aligned} &\varrho \frac{d\vec{V}}{dt} + \nabla_{\vec{r}} \cdot \left[\sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s\right] \\ &\qquad = \rho \vec{E} + \vec{J} \times \vec{B} - \nabla_{\vec{r}} \cdot \overleftrightarrow{P}^{CM} + \nabla_{\vec{r}} \cdot \left[\sum_s m_s n_s \vec{w}_s \otimes \vec{w}_s\right]. \end{aligned} \tag{A.4}$$

The divergence terms cancel exactly, yielding the single fluid momentum equation:

$$\varrho \frac{d\vec{V}}{dt} = \rho \vec{E} + \vec{J} \times \vec{B} - \nabla_{\vec{r}} \cdot \overleftrightarrow{P}^{CM}, \tag{A.5}$$

where the center-of-mass convective derivative is:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \vec{V} \cdot \nabla_{\vec{r}}.$$

A.3. Derivation and motivation of adiabatic equation of state

We show that for an ideal gas, conserving entropy along fluid trajectories is equivalent to the relation $\frac{d}{dt}(P\varrho^{-\gamma}) = 0$.

First Law of Thermodynamics

Per unit mass, the first law reads

$$de = T\,ds - P\,d\left(\frac{1}{\varrho}\right), \tag{A.6}$$

where e is specific internal energy (per unit mass), s is specific entropy (per unit mass), $1/\varrho = 1/nm$ is specific volume (per unit mass), n is number density and m is mass per particle.

Ideal Gas Relations

For an ideal gas two relations hold:

$$e = c_v T, \tag{A.7}$$

$$P = \frac{k_B}{m}\,\varrho T, \tag{A.8}$$

where c_v is the specific heat at constant volume, k_B is Boltzmann's constant, and m is the particle mass.

Differential of Entropy

Substituting (A.7) and (A.8) into (A.6) gives

$$ds = c_v\,\frac{dT}{T} - \frac{k_B}{m}\,\frac{d\varrho}{\varrho}. \tag{A.9}$$

Integrating:

$$s = c_v \ln T - \frac{k_B}{m} \ln \varrho + \text{const.} \tag{A.10}$$

Eliminating Temperature

From (A.8), $T = Pm/(\varrho k_B)$, so $\ln T = \ln P - \ln \varrho + \text{const.}$ Substituting into (A.10):

$$s = c_v \ln\left(\frac{P}{\varrho}\right) - \frac{k_B}{m} \ln \varrho + \text{const.}$$

Expanding and collecting the $\ln \varrho$ terms:

$$s = c_v \ln P - \left(c_v + \frac{k_B}{m} \right) \ln \varrho + \text{const.}$$

Using the ideal-gas identity $c_p - c_v = k_B/m$ and the definition $\gamma \equiv c_p/c_v$, we have $c_v + k_B/m = c_p = \gamma c_v$, so

$$s = c_v(\ln P - \gamma \ln \varrho) + \text{const} = c_v \ln\bigl(P \varrho^{-\gamma}\bigr) + \text{const.} \tag{A.11}$$

Adiabatic Condition

Since $c_v > 0$, equation (A.11) shows that

$$\frac{ds}{dt} = 0 \quad \Longleftrightarrow \quad \frac{d}{dt}\bigl(P \varrho^{-\gamma}\bigr) = 0,$$

where $d/dt = \partial_t + \mathbf{v} \cdot \nabla$ is the material derivative following a fluid element. This is the *adiabatic equation of state*, it simply states that entropy is conserved along fluid trajectories.

Note: Plasma is not an ideal gas

The derivation relies on the ideal gas law (A.8). In a plasma context this assumption should be revisited, as inter-particle interactions and anisotropic pressure can modify the equation of state.